

Identification of dynamical systems

John Lataire

Contents

Organisation	iii
About this course	iii
Timeline	iv
1 Preliminaries	1
1.1 Expected background knowledge	1
1.2 Fourier spectra and DFT	1
1.3 Auto and cross power spectra	1
2 Non-parametric identification methods	2
2.1 Block structures and noise assumptions	2
2.2 Estimation of noise (co-)variances	3
2.2.1 Variance of the sample mean	5
2.3 Maximum Likelihood FRF estimation	5
2.3.1 Approximate uncertainty of the FRF based on the coherence	7
2.4 Detection and Quantification of nonlinear distortions	8
2.4.1 Response to a single tone	8
2.4.2 Response to a multisine excitation	11
2.4.3 The ‘fast’ method	12
2.4.4 The best linear approximation	14
2.4.5 The ‘robust’ method	18
2.5 The BLA in feedback conditions	21
2.5.1 1st order estimation of a nonlinearity in closed loop	26
2.6 Detection and quantification of time-varying systems	28
3 Parametric models of LTI systems	29
3.1 Band-limited vs Zero-Order-Hold setups	29
3.1.1 Band-limited and ZOH noise models	31
3.2 Lumped versus distributed systems	34
3.3 A generalised model structure and frequency variable Ω	37
3.3.1 Parametrisation of rational forms	38
4 Parametric transfer function identification of LTI systems	42
4.1 Maximum likelihood estimation, with known non-parametric noise model	42
4.1.1 Approximate maximum likelihood as initial estimates	45
LLS (Linear least squares)	46
IWLS (Iterative Weighted Least Squares)	46
IQML (Iterative Quadratic Maximum Likelihood)	46
TLS (Total Least Squares)	47
GTLS (Generalized Total Least Squares)	48
BTLS (Bootstrapped Total Least Squares)	48
4.1.2 Imposing real parameter values	49

4.1.3	Computing the maximum likelihood estimate	52
4.1.4	Computing the covariance of the maximum likelihood estimates	54
4.1.5	Covariance of the estimated poles and zeroes	54
4.1.6	Model structure and complexity selection, and model validation	55
4.2	Simultaneous noise and plant model estimation	56
4.3	Taking into account the setpoint of the system	59
5	Improved non-parametric methods	60
5.1	Local parametric methods	60
5.1.1	Local polynomial model	61
5.1.2	Local rational model	61
6	Guidelines	62
6.1	Basic questions: Stochastic framework and Application	62
6.2	Identification step by step	63
	Bibliography	68

Organisation

About this course

This course builds on the concepts and tools acquired in the 1st Master course on *Measurements and Data Driven Modeling* to obtain parametric models of dynamical systems based on measured data.

You will gain experience and become aware of pitfalls when extracting (non-) parametric models of dynamic systems from measured (or simulated) data. This encompasses the detection and quantification of disturbances (noise, nonlinear distortions and time variations), the characterisation of the measurement setup, the selection of an appropriate model structure, the optimisation of the model complexity, the validation of the estimated model, and obtaining the uncertainty of the estimated model parameters.

Since, the best way to learn the above tools is to apply them yourself, an important part of the course will consist of a project, where you will be in charge of performing the measurements on a dynamic system and to extract a parametric model of it.

[ECTS fiche](#)

Timeline

This timeline is **tentative**, and will be updated as the lectures and exercise sessions progress. *Based on academic year 2024-2025*

Session 1 (September 17, 2025).

- Welcome to the course.
- Organisation and expectations:
 - Lectures and project, see [Canvas](#)
- Overview of the material, see the table of contents on page [ii](#).
- Check schedule, adapt to availabilities (internships?).
- Section 2.1: [Block structures and noise assumptions](#) (p 2). Block structures and noise assumptions, non-parametric. Periodic versus arbitrary, open loop vs closed loop.
- Section 2.2: [Estimation of noise \(co-\)variances](#) (p 3). Non-parametric FRF estimation: determine the variance of the FRF estimate based on the variance of the spectra.
- Section 2.3.1: [Approximate uncertainty of the FRF based on the coherence](#) (p 7), for arbitrary excitations.
- Section 2.4: [Detection and Quantification of nonlinear distortions](#) (p 8). Non-linear distortions, explained on static nonlinear systems. Distinction between even and odd nonlinear distortions by using odd multisines.

Session 2 (September 22, 2025). Prepare exercises from [4], available on the [Canvas Exercises](#) page.

- Non-parametric estimation of FRFs, influence of the noise. Exercise numbers 43, 44, 46, 47.
- Discussion exercises
 - 43, 44: FRF measurement, output noise only, random and multisine excitation: determine the variance of the estimate theoretically and via sample variances. Make the plots of the estimates, the estimated variance based on the sample variance of the spectra, and the actual sample variance of the estimated FRF.
 - 46, 47: FRF measurements with input AND output noise: observe the deterministic error in the case of random noise excitation (from spectral averaging technique), which is not there for the multisine excitation

Session 3 (September 24, 2025).

- Continue with the exercises from Session 2.
- Exercises in [4] on the detection of nonlinear distortions:
 - 77 a) single sine, b) multisine response of a static NL system

Session 4 (September 29, 2025).

- Discussion on exercises
- Section 2.4: [Detection and Quantification of nonlinear distortions](#) (p 8)
- The ‘fast’ and the ‘robust’ method:
 - Section 2.4.3: [The ‘fast’ method](#) (p 12)
 - Section 2.4.5: [The ‘robust’ method](#) (p 18)
- About coherent and stochastic nonlinear distortions
 - BLA: (true) underlying linear system + bias term (coherent nonlinear contributions)
 - stochastic nonlinear distortions y_s (non-coherent nonlinear contributions)

Session 5 (October 6, 2025).

- Best Linear Approximation and (stochastic nonlinear distortions), recap of Section 2.4: [Detection and Quantification of nonlinear distortions](#) (p 8).
- Section 2.5: [The BLA in feedback conditions](#) (p 21)
- Start with Chapter 3: [Parametric models of LTI systems](#) (p 29)
 - Section 3.1: [Band-limited vs Zero-Order-Hold setups](#) (p 29)

Session 6 (October 8, 2025).

- Follow up on earlier exercises
- Starting with exercises on dynamic NL systems
 - 80 multisine response of a dynamic nonlinear system
 - 81 quantification of nonlinear distortions with multisine excitation

Session 7 (October 13, 2025).

- Continue with parametric models of LTI systems
 - Section 3.1.1: [Band-limited and ZOH noise models](#) (p 31)
 - Section 3.2: [Lumped versus distributed systems](#) (p 34)
 - Section 3.3: [A generalised model structure and frequency variable \$\Omega\$](#) (p 37)
 - Section 3.3.1: [Parametrisation of rational forms](#) (p 38)
- Start with Chapter 4: [Parametric transfer function identification of LTI systems](#) (p 42)
 - Section 4.1: [Maximum likelihood estimation, with known non-parametric noise model](#) (p 42)

⇓ ————— up-to-date until here ————— ⇓

Session 8 (October 15, 2025). Continue with exercises.

Session 9 (October 20, 2025). Parametric transfer function identification

- Started with Section 4.1.1: [Approximate maximum likelihood as initial estimates](#) (p 45): Linear least squares (dependence on the constraint used, importance of frequency weight, TLS implementation with SVD, IWLS (Sanathanan Koerner if weighting with $|A|^2$)).

Maximum likelihood, parametric estimation of LTI transfer functions

- Section 4.1.1: [Approximate maximum likelihood as initial estimates](#) (p 45)
 - IQML Iterative Quadratic Maximum Likelihood
 - GTLS, implementation with the GSVD
 - BTLS (Bootstrapped Total Least Squares)
- Section 4.1.3: [Computing the maximum likelihood estimate](#) (p 52) with Gauss-Newton and the Levenberg-Marquardt optimisations.

Session 10 (October 23, 2024). Start of the project

- Expectations for the project
- Introduction to the VXI measurement system
- Continuing the exercises

Session 11 (December 4, 2024).

- Section 4.1.2: [Imposing real parameter values](#) (p 49), including how to compute $C_{J_{re}}$ (i.e. the column covariance matrix of J_{re}).

- Section 4.1.6: *Model structure and complexity selection, and model validation* (p 55)
- Normalisation of the regressor matrix and the Jacobian, as mentioned in the slide on page 52.
- Section 4.1.4: *Computing the covariance of the maximum likelihood estimates* (p 54), and Section 4.1.5: *Covariance of the estimated poles and zeroes* (p 54).

Not seen in 2425

- Improved non-parametric (Local parametric) identification
- Exercises on the Best Linear Approximation, from [4].
 - Robust method: 88a
 - Fast method: 89 abc (long exercise)

Preliminaries

1.1 Expected background knowledge

System theory Impulse response, Transfer function, Pole-zero representations and interpretations

Spectral representations of signals Laplace transform, Fourier transform.

Digital signal processing Discrete Fourier Transform (DFT), bin (normalised frequency unit), frequency resolution

Frequency Response Function (FRF) measurements Steady-state vs transient response, windowing

Statistics Random noise, first and second moments, probability density distribution, stochastic convergence (law of large numbers)

Common estimators (Weighted) (Non-)linear Least Squares, Maximum Likelihood

Estimator properties Bias, variance, consistency, efficiency

1.2 Fourier spectra and DFT

Definition of the Discrete Fourier Transform

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \quad (1.1)$$

1.3 Auto and cross power spectra

Definitions of the auto-power spectrum S_{XX} and cross-power spectrum S_{YX} .

$$S_{YX} = \mathbb{E} \{ Y \bar{X} \}, \quad \hat{S}_{YX} = \frac{1}{M} \sum_{l=1}^M Y^{[l]} \overline{X^{[l]}} \quad (1.2)$$

where the estimate $\hat{S}_{YX}$ is an average over M realisations ($X^{[l]}$ is the l th realisation of X).

Non-parametric identification methods

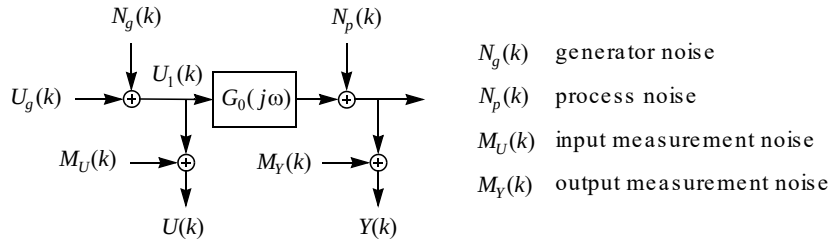
2.1 Block structures and noise assumptions

Consider the setup with generator noise, process noise, and measurement noise. We make a distinction between arbitrary and periodic excitations because parts of the noise act as an excitation (namely N_g) for the arbitrary case, while it acts as a disturbance in the periodic case.

Also, recall that an arbitrary excitation results in a transient term T_G . Periodic signals are considered in steady-state condition.

1. Nonparametric methods for LTI systems

1.a Problem statement



Noisy observations

$$Y(k) = Y_0(k) + N_Y(k)$$

$$U(k) = U_0(k) + N_U(k)$$

- periodic signals

$$\begin{cases} Y_0(k) = G_0(j\omega_k)U_0(k) \\ U_0(k) = U_g(k) \end{cases} \text{ and } \begin{cases} N_Y(k) = G_0(j\omega_k)N_g(k) + N_p(k) + M_Y(k) \\ N_U(k) = N_g(k) + M_U(k) \end{cases}$$

- arbitrary signals

$$\begin{cases} Y_0(k) = G_0(j\omega_k)U_0(k) + T_G(j\omega_k) \\ U_0(k) = U_1(k) \end{cases} \text{ and } \begin{cases} N_Y(k) = N_p(k) + M_Y(k) \\ N_U(k) = M_U(k) \end{cases}$$

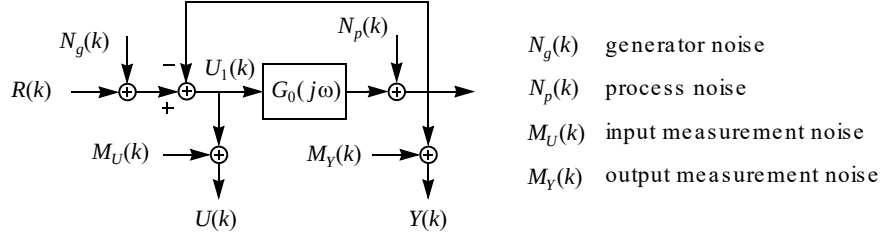
5/316

For the closed loop situation, the reasoning is adapted. Denote again

$$Y(k) = Y_0(k) + N_y(k), \quad U(k) = U_0(k) + N_u(k), \quad (2.1)$$

1. Nonparametric methods for LTI systems

1.a Problem statement (cont'd)



- periodic signals

$$\begin{cases} Y_0(k) = G_0(j\omega_k)U_0(k) \\ U_0(k) = \frac{1}{1 + G_0(j\omega_k)}R(k) \end{cases} \text{ and } \begin{cases} N_Y(k) = \frac{G_0(j\omega_k)N_g(k) + N_p(k)}{1 + G_0(j\omega_k)} + M_Y(k) \\ N_U(k) = \frac{N_g(k) - N_p(k)}{1 + G_0(j\omega_k)} + M_U(k) \end{cases}$$

- arbitrary signals

$$\begin{cases} Y_0(k) = G_0(j\omega_k)U_0(k) + T_G(j\omega_k) \\ U_0(k) = U_1(k) \end{cases} \text{ and } \begin{cases} N_Y(k) = N_p(k) + M_Y(k) \\ N_U(k) = M_U(k) \end{cases}$$

6/316

For the closed loop case, observe that the noiseless and noisy signals depend on whether the *periodic* or the *arbitrary* case is considered. Specifically, the process noise N_p only appears in the noise for the periodic case, while for the arbitrary case, N_p is also a part of the excitation (it affects U_1 via the feedback loop). Namely,

$$U_1(k) = \frac{R(k) - N_p(k) + N_g(k)}{1 + G_0(j\omega_k)}. \quad (2.2)$$

As such, N_y is correlated with the excitation U_1 .

Exercise 2.1. Is it possible to identify G in the case of arbitrary signals if only U and Y are given? What about the case of periodic signals?

2.2 Estimation of noise (co-)variances

First recall the definition of the first and second moments of the (complex) noise and the property of circular complex distributed noise.

1 Nonparametric methods for LTI systems

1.a Problem statement (cont'd)

- Nonparametric representation

Frequency response function

$$G_0(j\omega_k), k = 1, 2, \dots$$

(Co-)variances circular complex distributed noise

$$\sigma_Y^2(k) = \text{var}(N_Y(k)) = E\{|N_Y(k)|^2\}$$

$$\sigma_U^2(k) = \text{var}(N_U(k)) = E\{|N_U(k)|^2\}, k = 1, 2, \dots$$

$$\sigma_{YU}^2(k) = \text{covar}(N_Y(k), N_U(k)) = E\{N_Y(k)\overline{N_U(k)}\}$$

and

$$E\{N_Y^2(k)\} = 0$$

$$E\{N_U^2(k)\} = 0$$

$$E\{N_Y(k)N_U(k)\} = 0$$

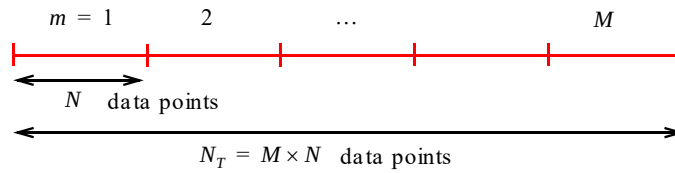
8/316

When working with periodic excitations, the sample means and sample variances of the input and output spectra can be obtained by averaging over consecutive periods of the steady-state response.

1. Nonparametric methods for LTI systems

1.b Periodic excitations – errors-in-variables problem

- Noisy input-output measurements
- Observe M consecutive periods of the steady state response



Sample means

$$\hat{Y}(k) = \frac{1}{M} \sum_{m=1}^M Y^{[m]}(k), \hat{U}(k) = \frac{1}{M} \sum_{m=1}^M U^{[m]}(k)$$

Sample (co-)variances

$$\hat{\sigma}_Y^2(k) = \frac{1}{M(M-1)} \sum_{m=1}^M |Y^{[m]}(k) - \hat{Y}(k)|^2, \hat{\sigma}_U^2(k) = \frac{1}{M(M-1)} \sum_{m=1}^M |U^{[m]}(k) - \hat{U}(k)|^2,$$

$$\hat{\sigma}_{YU}^2(k) = \frac{1}{M(M-1)} \sum_{m=1}^M (Y^{[m]}(k) - \hat{Y}(k))(\overline{U^{[m]}(k) - \hat{U}(k)})$$

9/316

Note the division by $\frac{1}{M(M-1)}$ for computing the sample (co-)variances. The factor $\frac{1}{M-1}$ is the (usual) normalisation to estimate the variance. The additional factor $\frac{1}{M}$ is because we consider the variance of the *sample mean* (i.e. the variance of $\hat{Y}$, not of Y), which decreases proportionally to M .

1. Nonparametric methods for LTI systems

1.b Periodic excitations – errors-in-variables problem (cont'd)

• Maximum likelihood estimate frequency response function

$Z^{[m]} = [Y^{[m]}(k), U^{[m]}(k)]^T$ is independent (over m), circular complex normally distributed $\Rightarrow$
Gaussian probability density function of the measurements Z

$$\begin{aligned} f_Z(Z|\theta) &= \prod_{m=1}^M f_{Z^{[m]}|Z}(Z^{[m]}|\theta) \\ &= \frac{1}{\pi^{2M} \prod_{m=1}^M \det(C_Z)} \exp\left(-\sum_{m=1}^M (Z^{[m]} - Z_p)^H C_Z^{-1} (Z^{[m]} - Z_p)\right) \end{aligned}$$

with

$$\theta = [Z_p^T, G(j\omega_k)]^T, Z_p = \begin{bmatrix} Y_p(k) \\ U_p(k) \end{bmatrix} \text{ and } C_Z = \text{Cov}(Z^{[m]}) = \begin{bmatrix} \sigma_Y^2(k) & \sigma_{YU}(k) \\ \sigma_{YU}(k) & \sigma_U^2(k) \end{bmatrix}$$

Gaussian ML cost function

$$V_{\text{ML}}(G(j\omega_k), Z_p, Z) = \sum_{m=1}^M (Z^{[m]} - Z_p)^H C_Z^{-1} (Z^{[m]} - Z_p) + \lambda[1, -G(j\omega_k)] Z_p$$

Eliminate $Z_p, \lambda \Rightarrow$

$$\hat{G}_{\text{ML}}(j\omega_k) = \frac{\hat{Y}(k)}{\hat{U}(k)} = \frac{M^{-1} \sum_{m=1}^M Y^{[m]}(k)}{M^{-1} \sum_{m=1}^M U^{[m]}(k)}$$

10/316

The result for $\hat{G}_{\text{ML}}$ is also true for non-diagonal C_Z .

Proof. Compute the derivatives of the Lagrangian $\mathcal{L} \triangleq V_{\text{ML}}$ w.r.t. G, Z_p and λ , and set to 0.

$$\frac{\partial \mathcal{L}}{\partial Z_p} = 0 = -\sum_m (Z^{[m]} - Z_p)^H C_Z^{-1} + \lambda[1, -G] \quad (2.6)$$

$$\frac{\partial \mathcal{L}}{\partial G} = 0 = \lambda[0, -1] Z_p \quad \Rightarrow \lambda = 0 \quad (2.7)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0 = [1, -G] Z_p \quad \Rightarrow Y_p = G U_p \quad (2.8)$$

□

Plugging (2.7) and (2.8) into (2.6) (eliminating Y_p and λ) gives

$$\sum_m [Y^{[m]}, U^{[m]}] C_Z^{-1} - U_p [G, 1] C_Z^{-1} = 0 \quad (2.9)$$

Right-multiply by C_Z and write both equations explicitly:

$$\sum_m (Y^{[m]} - G U_p) = 0 \quad \sum_m (U^{[m]} - U_p) = 0, \quad (2.10)$$

from which $U_p = \frac{1}{M} \sum_{m=1}^M U^{[m]}$ and $G = \frac{\frac{1}{M} \sum_{m=1}^M Y^{[m]}}{\frac{1}{M} \sum_{m=1}^M U^{[m]}}$.

1. Nonparametric methods for LTI systems

1.b Periodic excitations – errors-in-variables problem (cont'd)

- Maximum likelihood estimate frequency response function (cont'd)

Relative bias on FRF estimate

$$b(k) = \frac{E\{\hat{G}(j\omega_k)\}}{G_0(j\omega_k)} - 1 = -\exp\left(-\frac{|U_0(k)|^2}{\sigma_U^2(k)}\right) \left(1 - \frac{\sigma_{YU}^2(k)}{\sigma_U^2(k)\sigma_Y^2(k)} \frac{U_0(k)/\sigma_U(k)}{Y_0(k)/\sigma_Y(k)}\right)$$

where

$$|b(k)| < 1 \times 10^{-4} \text{ if } \min(|U_0(k)|/\sigma_U(k), |Y_0(k)|/\sigma_Y(k)) > 10 \text{ dB}$$

Asymptotic ($M \rightarrow \infty$) variance

$$\hat{G}_{ML}(j\omega_k) \approx G_0(j\omega_k) + \delta_G(k)$$

with

$$\begin{aligned} \delta_G(k) &= G_0(j\omega_k) \left(\frac{1}{M} \sum_{m=1}^M \frac{N_Y^{[m]}(k)}{Y_0(k)} - \frac{1}{M} \sum_{m=1}^M \frac{N_U^{[m]}(k)}{U_0(k)} \right) \\ \text{var}(\delta_G(k)) &= |G_0(j\omega_k)|^2 \left(\frac{\sigma_Y^2(k)}{|Y_0(k)|^2} + \frac{\sigma_U^2(k)}{|U_0(k)|^2} - 2\text{Re}\left(\frac{\sigma_{YU}^2(k)}{Y_0(k)\overline{U_0(k)}}\right) \right) \end{aligned}$$

12/316

See Section 2.4.2.1 in [2] about the relative bias.

2.3.1 Approximate uncertainty of the FRF based on the coherence

If σ_Y^2 , σ_U^2 and σ_{YU}^2 are not known, the coherence can be used to estimate the variance of the FRF. Specifically, the coherence γ^2 is defined as

$$\gamma^2(\omega) = \frac{|S_{yu}|^2}{S_{uu}S_{yy}}, \quad (2.11)$$

where the required (cross-)power spectra are estimated from the measured DFT spectra. We can then show that the variance of the estimated FRF can be approximated as:

$$\sigma_G^2 \approx |\hat{G}|^2 \frac{1 - \hat{\gamma}^2}{\hat{\gamma}^2} \quad (2.12)$$

(See also [2, Section 2.5.4].)

Covariance SA, based on coherence (Section 2.5.4. in Pintelon)

$$\hat{d}^2 = \frac{|S_{YU}|^2}{S_{UU} S_{YY}}, \quad \hat{d}^2 = \frac{|1 + \frac{\sigma_{YU}^2}{|Y_0|^2}|^2}{\left(1 + \frac{\sigma_U^2}{|U_0|^2}\right) \left(1 + \frac{\sigma_Y^2}{|Y_0|^2}\right)} \approx \frac{|1 + \frac{\sigma_{YU}^2}{Y_0 U_0}|^2}{\left(1 + \frac{\sigma_U^2}{|U_0|^2} + \frac{\sigma_Y^2}{|Y_0|^2}\right)}$$

based on measured DFT

$$Y = Y_0 + N_Y, \quad Y_0 = G U_0$$

$$U = U_0 + N_U$$

$$\sigma_{YU}^2 \approx \frac{\hat{\sigma}_U^2 \hat{\sigma}_Y^2}{|U_0|^2 |Y_0|^2} \ll \frac{\hat{\sigma}_U^2}{|U_0|^2} + \frac{\hat{\sigma}_Y^2}{|Y_0|^2}$$

$$\frac{\hat{\sigma}_{YU}^2}{Y_0 U_0} \ll 1$$

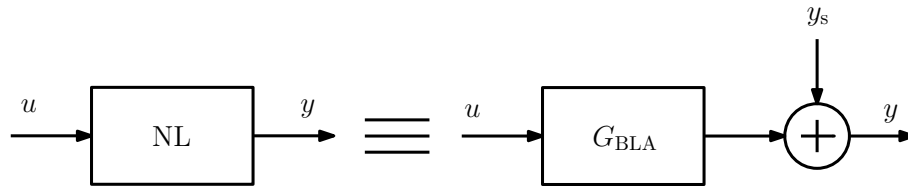
$$\hat{\sigma}_G^2 \approx |\hat{G}|^2 \frac{1 - \hat{d}^2}{\hat{d}^2} \approx |\hat{G}|^2 \left(\frac{\sigma_U^2}{|U_0|^2} - \frac{2\sigma_{YU}^2}{Y_0 U_0} + \frac{\sigma_Y^2}{|Y_0|^2} \right)$$

Useful when variances cannot be estimated from the data

$$\frac{1 - \hat{d}^2}{\hat{d}^2} = \frac{1 - \frac{|1 + \frac{\sigma_{YU}^2}{Y_0 U_0}|^2}{\left(1 + \frac{\sigma_U^2}{|U_0|^2} + \frac{\sigma_Y^2}{|Y_0|^2}\right)}}{\frac{|1 + \frac{\sigma_{YU}^2}{Y_0 U_0}|^2}{\left(1 + \frac{\sigma_U^2}{|U_0|^2} + \frac{\sigma_Y^2}{|Y_0|^2}\right)}} = \frac{\frac{\sigma_U^2}{|U_0|^2} + \frac{\sigma_Y^2}{|Y_0|^2} - \frac{2\sigma_{YU}^2}{Y_0 U_0}}{|1 + \frac{\sigma_{YU}^2}{Y_0 U_0}|^2}$$

2.4 Detection and Quantification of nonlinear distortions

The goal of this section is to describe a nonlinear system as a Best Linear Approximation (BLA) and a source of distortion y_s on the output.



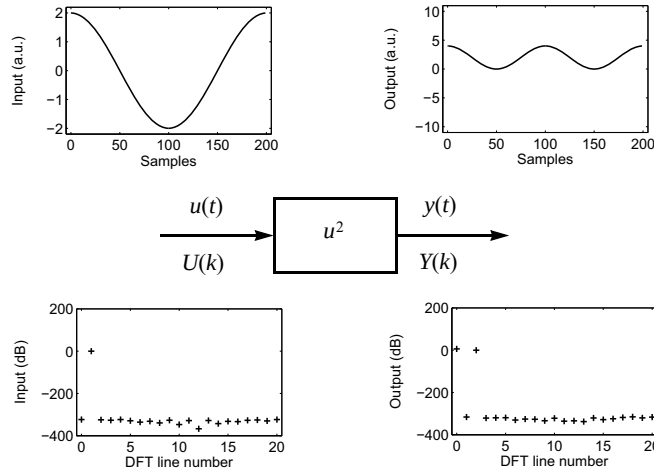
Consider nonlinear systems which satisfy the PISPO property (Periodic Input Same Period Out). These are systems which, when excited by a periodic input signal, have a periodic output with the same period as the input signal. This allows for systems with saturation and discontinuities. However, chaotic systems, or systems generating subharmonics are not included.

2.4.1 Response to a single tone

When excited by a single tone, these systems will generate harmonics in their response, as illustrated here.

1. Nonparametric methods for LTI systems

1.f Nonlinear systems – response to a periodic input



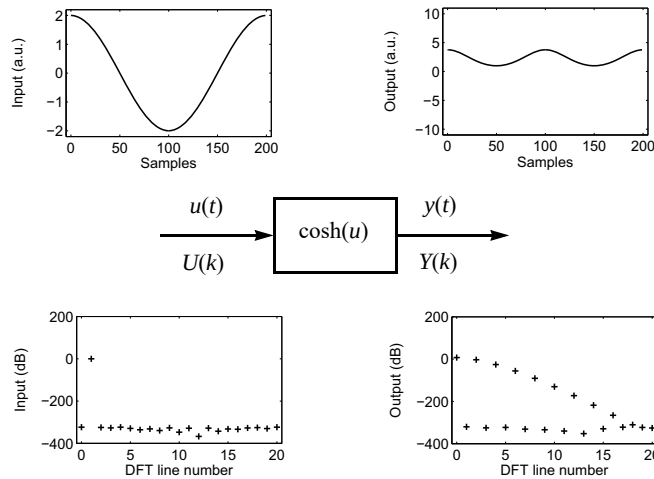
$$\cos^2(\omega_0 t) = \left(\frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t}) \right)^2 = \frac{1}{4}(e^{j2\omega_0 t} + 2 + e^{-j2\omega_0 t}) = \frac{1}{2}(1 + \cos(2\omega_0 t))$$

20/316

Important observation for this quadratic system: a single sine with frequency ω_0 generates a response at 2 frequencies, namely at 0 and at $2\omega_0$.

1. Nonparametric methods for LTI systems

1.f Nonlinear systems – response to a periodic input (cont'd)



$$\cos^{2n}(\omega_0 t) = \frac{1}{2^{2n}} C_{2n}^n + \frac{1}{2^{2n-1}} \sum_{r=0}^{n-1} C_n^r \cos(2(n-r)\omega_0 t)$$

21/316

Observe that the system $y = \cosh(u)$, which is an *even* system will only generate power at $2m\omega_0$, with $m \in \mathbb{Z}$, i.e. at even multiples of the input frequency ω_0 .

We have the following definitions:

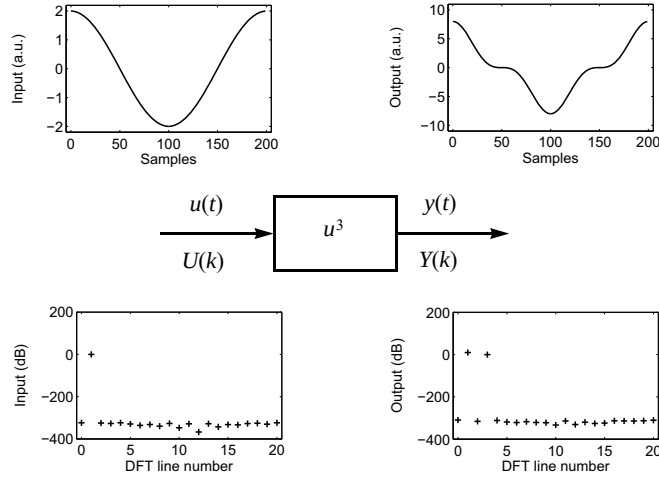
Even system: $G\{-u\} = G\{u\}$

Odd system: $G\{-u\} = -G\{u\}$

We can make a similar reasoning for *odd* systems, below.

1. Nonparametric methods for LTI systems

1.f Nonlinear systems – response to a periodic input (cont'd)

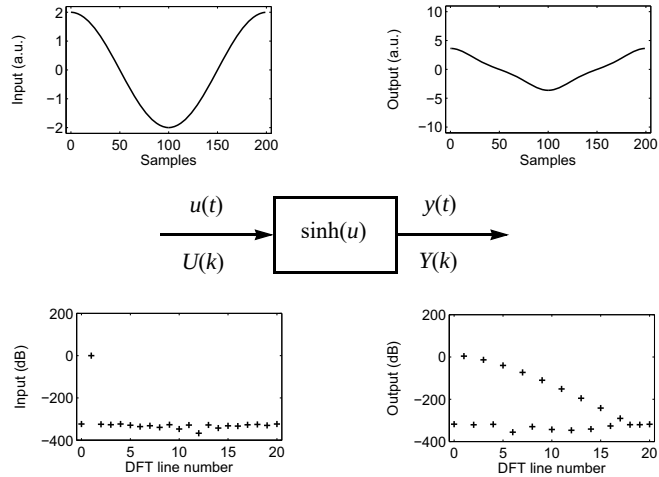


$$\cos^3(\omega_0 t) = \frac{1}{8}(e^{j3\omega_0 t} + 3e^{j\omega_0 t} + 3e^{-j\omega_0 t} + e^{-j3\omega_0 t}) = \frac{1}{4}(3\cos(\omega_0 t) + \cos(3\omega_0 t))$$

22/316

1. Nonparametric methods for LTI systems

1.f Nonlinear systems – response to a periodic input (cont'd)



$$\cos^{2n+1}(\omega_0 t) = \frac{1}{2^{2n}} \sum_{r=0}^n C_{2n+1}^r \cos((2(n-r)+1)\omega_0 t)$$

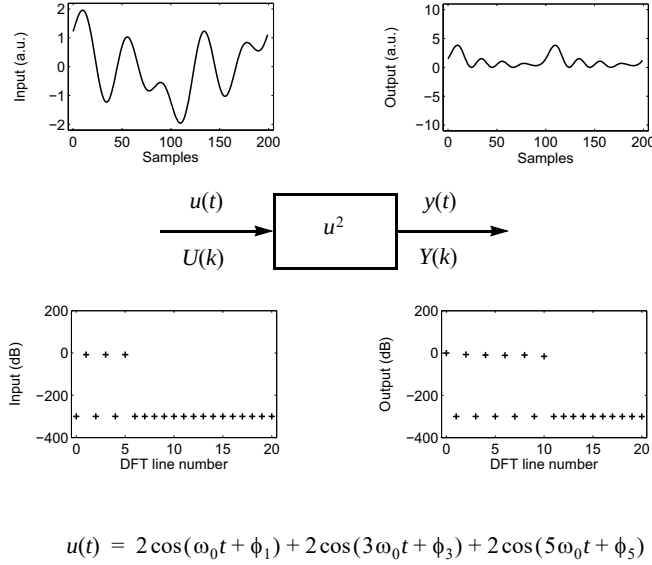
23/316

Namely, a single tone results in tones at odd multiples of the input frequency.

2.4.2 Response to a multisine excitation

1. Nonparametric methods for LTI systems

1.f Nonlinear systems – response to a periodic input (cont'd)

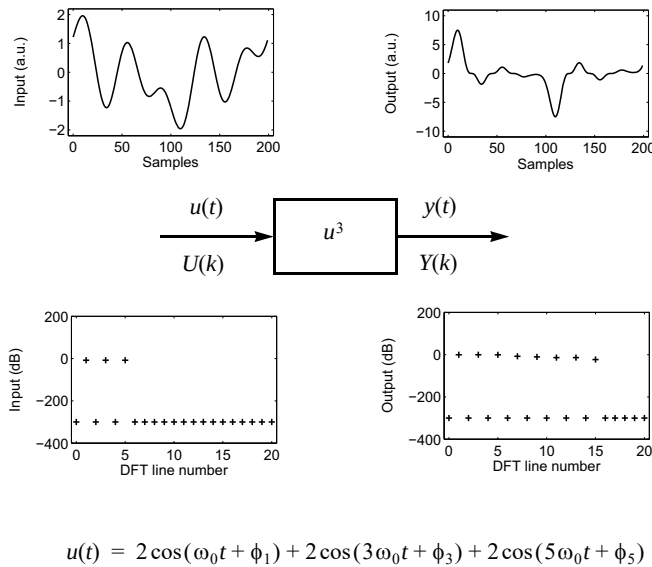


24/316

The response of an *even* system to a multisine consists of tones at *even* combinations of the input frequencies.

1. Nonparametric methods for LTI systems

1.f Nonlinear systems – response to a periodic input (cont'd)



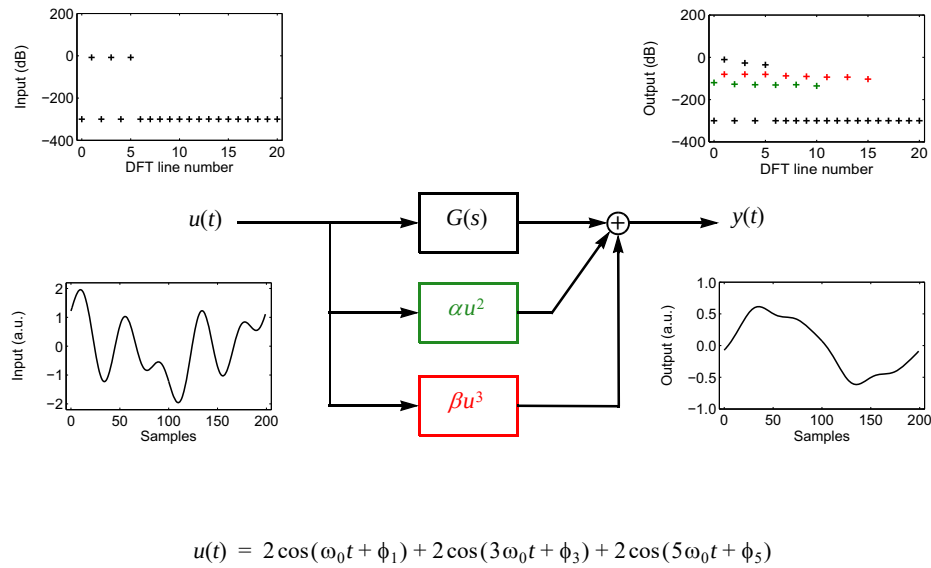
25/316

The response of an *odd* system to a multisine consists of tones at *odd* combinations of the input frequencies.

In practice, systems can be split into *even* and *odd* parts.

1. Nonparametric methods for LTI systems

1.f Nonlinear systems – response to a periodic input (cont'd)



26/316

2.4.3 The 'fast' method

Exercise 2.2. Demonstrate that the use of multisines which only excite odd multiples of the fundamental frequency will result in responses where

- the odd nonlinear contributions only appear at the odd output frequencies, and
- the even nonlinear contributions only appear at the even output frequencies.

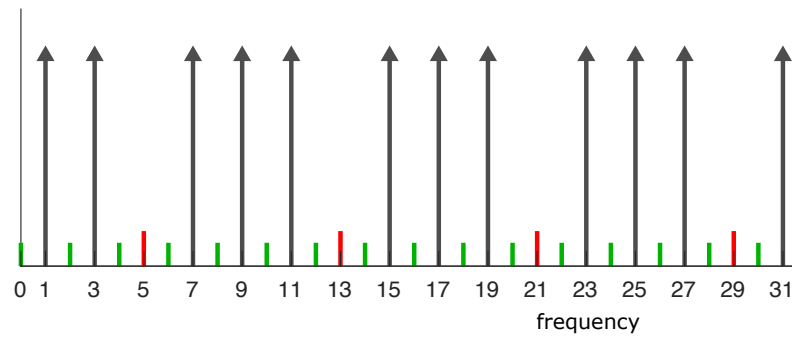
The important message of this exercise is that we can make a distinction between

- the (mainly) linear contributions,
- the odd nonlinear contributions,
- the even nonlinear contributions

by using a *single* well-designed experiment. Namely, the excitation signal should be a (random phase) multisine signal, which only excites odd multiples of the fundamental frequency. This approach is called 'fast' method for nonlinear distortion detection.

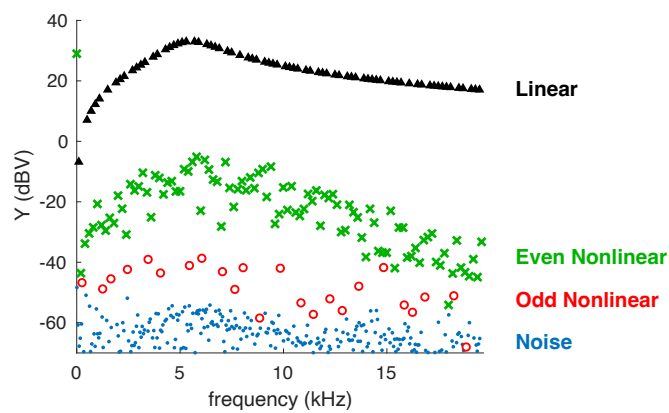
The detection of odd nonlinear contributions requires the omission of some odd frequencies in the excitation signal (red in the figure below). All even frequencies are used to detect the even nonlinear contributions (green in the figure below).

Multisines for Nonlinear Systems



14

Multisines for Nonlinear Systems



17

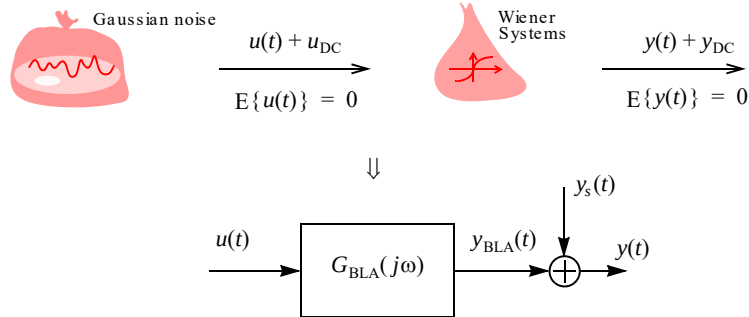
As visible above, the *noise* (in blue) can also be distinguished. This is done by measuring $P > 1$ periods of the excitation signal (in steady-state).

A more thorough explanation on nonlinear distortions is given in the course on [Selected Topics in Nonlinear System Identification](#) and in Chapter 4 of [2].

2.4.4 The best linear approximation

1. Nonparametric methods for LTI systems

1.g Nonlinear systems – best linear approximation (cont'd)



Definition best linear approximation (BLA):

$$G_{BLA}(j\omega) = \frac{S_{yu}(j\omega)}{S_{uu}(j\omega)} = \frac{F\{E\{y(t)u(t-\tau)\}\}}{F\{E\{u(t)u(t-\tau)\}\}}$$

 where the expected value is taken w.r.t. the random realisation of $u(t)$

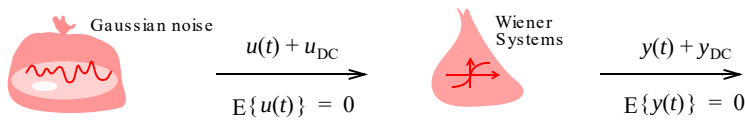
30/316

 where $F\{\cdot\}$ denotes the Fourier transform.

The best linear approximation of a nonlinear system depends on the *class of excitation signals* used. Two excitation signals are said to belong to the same class if they have the same *Riemann equivalent power spectrum*.

1. Nonparametric methods for LTI systems

1.g Nonlinear systems – best linear approximation


 Class of Gaussian excitation signals with the *same* Riemann equivalent power spectrum:

- Gaussian noise
- periodic Gaussian noise
- random phase multisine

$$u(t) = F^{-1/2} \sum_{k=-F, k \neq 0}^F U_k e^{j(2\pi k f_s / N)t}$$

with

 $F < N/2$, and F/N independent of N
 $|U_k|$ user defined, and $E\{e^{j\angle U_k}\} = 0$

 power $N^{-1} \sum_{n=0}^{N-1} u^2(nT_s)$ is independent of N

27/316

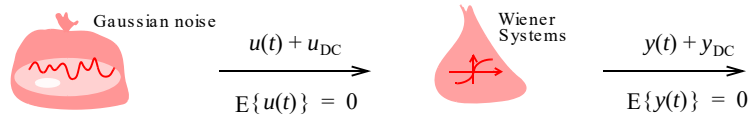
Note that U_k is a *complex* amplitude, such that it also includes the phase. These phases are chosen randomly, typically drawn from a uniform distribution

in $[0, 2\pi[$. From this, we have that $\mathbb{E}\{e^{j\angle U_k}\} = 0$.

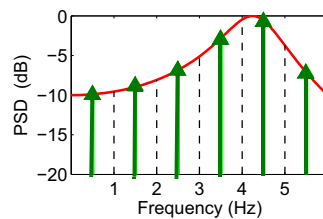
That the phases are chosen randomly will allow us to consider that the *nonlinear distortions* have a *stochastic* character.

1. Nonparametric methods for LTI systems

1.g Nonlinear systems – best linear approximation (cont'd)



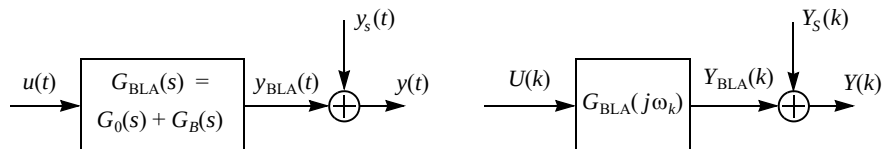
Gaussian noise and random phase multisines with the same Riemann equivalent power spectrum:



28/316

1. Nonparametric methods for LTI systems

1.g Nonlinear systems – best linear approximation (cont'd)



$G_0(s)$:

- true underlying linear system (if it exists)

$G_B(s)$:

- bias contribution
- depends on input power spectrum and odd nonlinear distortions only
- smooth function of the frequency

$G_{BLA}(s)$:

- best linear approximation or related linear dynamic system or LTI second order equivalent
- can be approximated very well by a rational form in s

31/316

The 'bias contribution' $G_B(s)$ is a contribution that is due to the nonlinear part of the system, but which generates components in the output spectrum that are *coherent* with the input spectrum. These coherent contributions can be generated by odd nonlinear contributions only.

For example, consider a *multisine* input signal u which excites the frequencies

ω_1 and ω_2 (with phases ϕ_1 and ϕ_2) and the system

$$y = u^3.$$

The spectral components in the (positive) output spectrum are

Coherent $\rightsquigarrow G_B$	Non-coherent $\rightsquigarrow y_s$
$e^{j(\omega_1+\omega_2-\omega_2)t+j(\phi_1+\phi_2-\phi_2)} = e^{j\omega_1 t+j\phi_1}$	$e^{j(2\omega_1+\omega_2)t+j(2\phi_1+\phi_2)}, e^{j(2\omega_1-\omega_2)t+j(2\phi_1-\phi_2)}$
$e^{j(\omega_2+\omega_1-\omega_1)t+j(\phi_2+\phi_1-\phi_1)} = e^{j\omega_2 t+j\phi_2}$	$e^{j(2\omega_2+\omega_1)t+j(2\phi_2+\phi_1)}, e^{j(2\omega_2-\omega_1)t+j(2\phi_2-\phi_1)}$
$e^{j(2\omega_1-\omega_1)t+j(2\phi_1-\phi_1)} = e^{j\omega_1 t+j\phi_1}$	$e^{j3\omega_1 t+j3\phi_1}, e^{j3\omega_2 t+j3\phi_2}$
$e^{j(2\omega_2-\omega_2)t+j(2\phi_2-\phi_2)} = e^{j\omega_2 t+j\phi_2}$	

It is clear that the *coherent contributions* appear at the frequencies which are excited in the input, and have a fixed phase difference with the corresponding spectral component of the input, thus appearing as a linear contribution, and are part of the response of G_B .

For the *non-coherent contributions* the phase relation is random w.r.t. the possible excited component in the input (remember that the phases of the input components are chosen randomly w.r.t. each other). These contributions are part of y_s .

Thus, the output spectrum of the nonlinear system can be written as

$$Y(k) = (G_0(j\omega_k) + G_B(j\omega_k))U(k) + Y_s(k), \quad (2.13)$$

giving, for the cross-power spectrum,

$$\begin{aligned} S_{YU}(k) &= \mathbb{E} \left\{ Y(k) \overline{U(k)} \right\} \\ &= (G_0(j\omega_k) + G_B(j\omega_k))S_{UU}(k) + \underbrace{\mathbb{E} \left\{ Y_s(k) \overline{U(k)} \right\}}_{=0} \end{aligned} \quad (2.14)$$

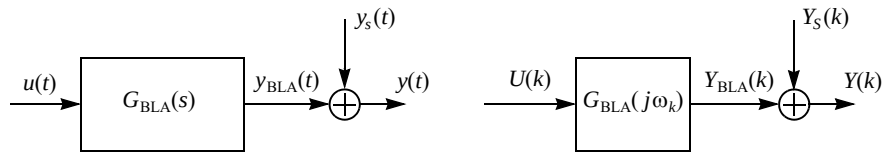
from which it is clear that

$$G_{\text{BLA}}(j\omega_k) \equiv \frac{S_{YU}}{S_{UU}} = G_0(j\omega_k) + G_B(j\omega_k). \quad (2.15)$$

In this course, we do not seek to make a distinction between G_0 and G_B . Only the sum, G_{BLA} , is of interest.

1. Nonparametric methods for LTI systems

1.g Nonlinear systems – best linear approximation (cont'd)



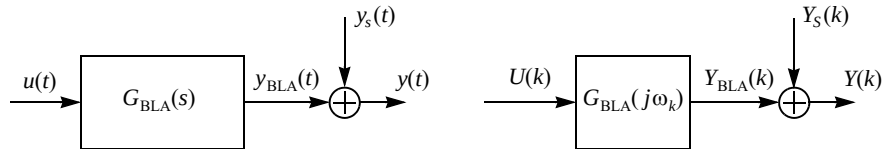
$y_s(t)$:

- periodic signal
- depends on input power spectrum and the even and odd nonlinearities
- zero mean
- uncorrelated with – but not independent of – $u(t)$ (Enqvist and Ljung, 2005)
- mostly not normally distributed

32/316

1. Nonparametric methods for LTI systems

1.g Nonlinear systems – best linear approximation (cont'd)



$Y_s(k)$:

- $E\{Y_s(k)\bar{U}(k)\} = 0 \Rightarrow Y_s(k)$ uncorrelated with – but *not* independent of – $U(k)$
- $E\{Y_s(k)\} = 0$
- $Y_s(k)$ is asymptotically uncorrelated over k (cumulant mixing over k)
- asymptotically normally distributed
- $E\{|Y_s(k)|^2\}$ is a smooth function of the frequency

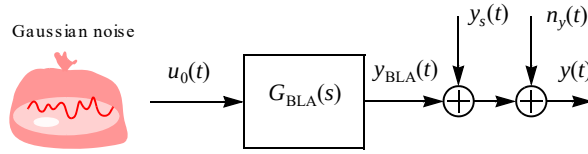
Note: expectations are taken over the different random realisations of the excitation

33/316

Asymptotically: number of excited frequencies $\rightarrow \infty$.

1. Nonparametric methods for LTI systems

1.h Nonlinear systems – noisy output measurements



$N_Y(k)$:

- independent of $U_0(k)$
- $E\{N_Y(k)\} = 0$
- asymptotically normally distributed
- asymptotically independent over k
- $E\{|N_Y(k)|^2\}$ is a smooth function of the frequency

Question

- distinction between $y_s(t)$ and $n_y(t)$?

34/316

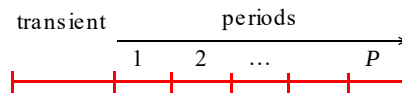
2.4.5 The 'robust' method

The distinction between $y_s(t)$ and $n_y(t)$ can be made thanks to the periodicity of $y_s(t)$ (same period as $u(t)$), whereas $n_y(t)$ is not periodic.

1. Nonparametric methods for LTI systems

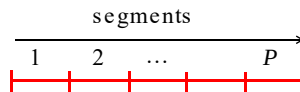
1.h Nonlinear systems – noisy output measurements (cont'd)

- FRF measurement using random phase multisine



$$U_0^{[p]} = U_0, Y_s^{[p]} = Y_s, N_Y^{[p]} \Rightarrow \begin{cases} \hat{G}_{ML} = G_{BLA} + Y_s/U_0 + \hat{N}_Y/U_0 \\ \text{var}(\hat{G}_{ML}) = \text{var}(N_Y^{[p]})/(P|U_0|^2) \end{cases}$$

- FRF measurement using Gaussian noise



$$U_0^{[p]}, Y_s^{[p]}, N_Y^{[p]} \Rightarrow \begin{cases} \hat{G}_{LS} = G_{BLA} + \hat{S}_{Y_s U_0} / \hat{S}_{U_0 U_0} + \hat{S}_{N_Y U_0} / \hat{S}_{U_0 U_0} \\ \text{var}(\hat{G}_{LS}) = (\text{var}(N_Y^{[p]}) + \text{var}(Y_s^{[p]})) / (P \hat{S}_{U_0 U_0}) \end{cases}$$

35/316

As is clear, for a single realisation of the excitation signal the variability of $\hat{G}$ is only due to the noise N_y in the periodic case, while it is due to both the noise and the stochastic nonlinear distortions in the random excitation (Gaussian noise)

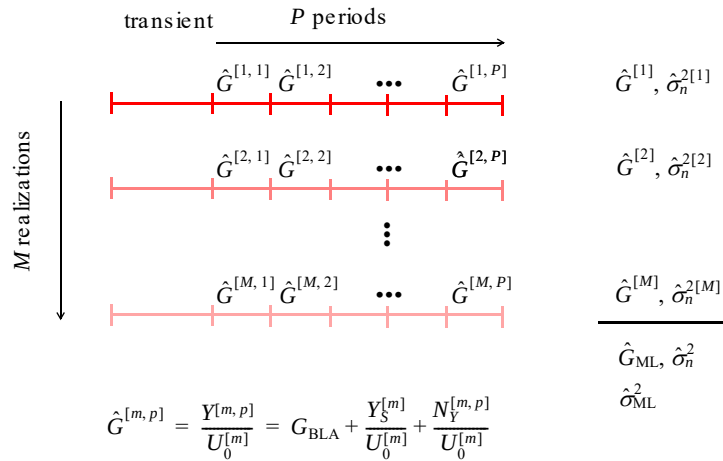
case. Therefore, a Gaussian noise excitation will *not* allow us to make a distinction between noise and nonlinear distortions.

A “robust” trick to distinguish the noise and nonlinear distortions by using a periodic excitation (multisine) will be to make *multiple experiments*, each time with a new *random phase realisation* of the multisine excitation.

1. Nonparametric methods for LTI systems

1.h Nonlinear systems – noisy output measurements (cont’d)

• FRF measurement using random phase multisine (cont’d)



36/316

Each random phase realisation (M in total) will result in a new realisation of the stochastic nonlinear distortions y_s . Note that the randomness of y_s relies on the randomness of the phases over the realisations.

The noise variance is determined via a variance computation *over the periods*,

The nonlinear distortion level is determined via a variance computation *over the realisations*.

To be more precise, the variance over the realisations will be affected by both the nonlinear distortions and the noise. However, since one can determine the variance due to the noise separately (over the periods), a distinction between both is made.

1. Nonparametric methods for LTI systems

1.h Nonlinear systems – noisy output measurements (cont'd)

- FRF measurement using random phase multisine (cont'd)

Sample mean and sample variance of each realisation

$$\hat{G}^{[m]} = \frac{1}{P} \sum_{p=1}^P \hat{G}^{[m,p]}$$

$$\hat{\sigma}_n^{2[m]} = \frac{1}{P(P-1)} \sum_{p=1}^P \left| \hat{G}^{[m,p]} - \hat{G}^{[m]} \right|^2, \quad \hat{\sigma}_n^2 = \frac{1}{M^2} \sum_{m=1}^M \hat{\sigma}_n^{2[m]}$$

Sample mean and sample variance of the mean values

$$\hat{G}_{\text{ML}} = \frac{1}{M} \sum_{m=1}^M \hat{G}^{[m]}$$

$$\hat{\sigma}_{\text{ML}}^2 = \frac{1}{M(M-1)} \sum_{m=1}^M |\hat{G}^{[m]} - \hat{G}_{\text{ML}}|^2$$

Properties

$$E\{\hat{\sigma}_n^2\} = \frac{\text{var}(N_Y)}{MP|U_0|^2}$$

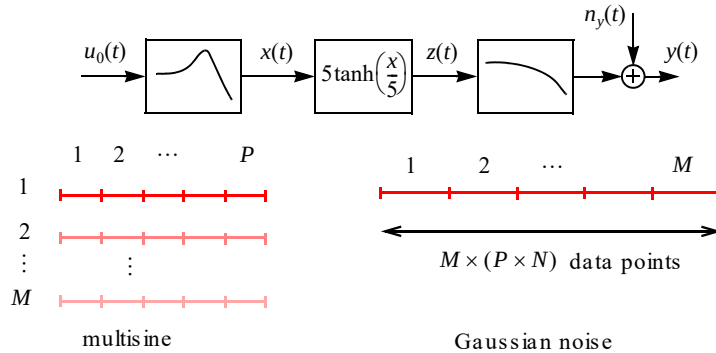
$$\mathbb{E}\{\hat{\sigma}_{\text{ML}}^2\} = \frac{\text{var}(Y_s)}{M|U_0|^2} + \frac{\text{var}(N_Y)}{MP|U_0|^2}$$

Note the asymmetric averaging of Y_s and N_Y

37/316

1. Nonparametric methods for LTI systems

1.i Simulation example: FRF estimate in the presence of nonlinear distortions



Multisine

- $M = 4, P = 2$, period length $N = 12500$
- $\text{std}(u(t)) = 1$

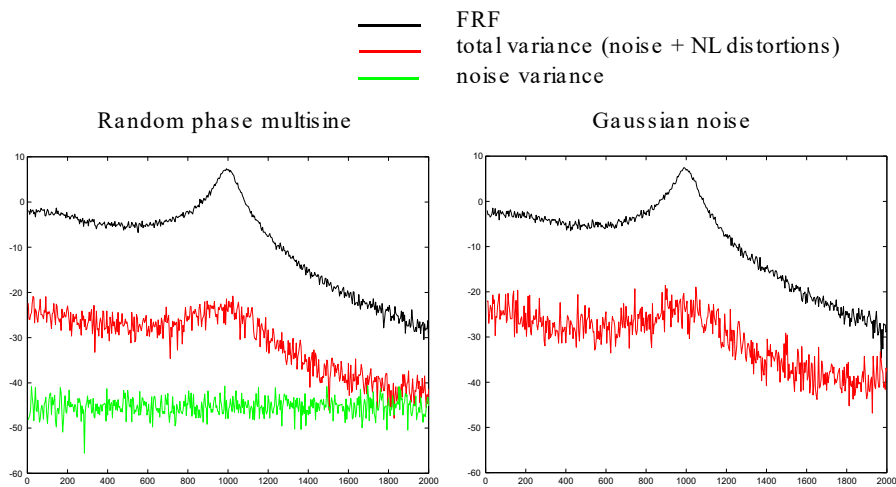
Gaussian noise

- record length $M \times P \times N \Rightarrow$ same measurement time
- segment length $N \times P \Rightarrow$ same averaging of the stochastic NL distortions
- same Riemann equivalent power spectrum

38/316

1. Nonparametric methods for LTI systems

1.i Simulation example: FRF estimate in the presence of nonlinear distortions (cont'd)



Note:

- random noise is $P = 2$ times more averaged for the multisine experiment

39/316

2.5 The BLA in feedback conditions

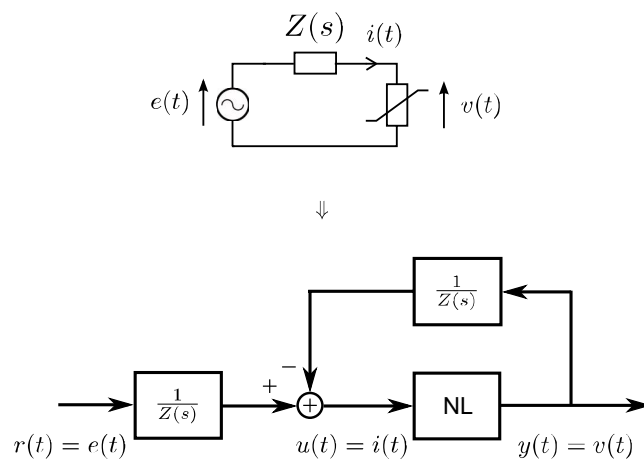
We would like to determine the FRF of the BLA of a system (plant) measured in feedback, in the case where both the plant and the feedback loop can be nonlinear.

Note that feedback can also be due to a non-ideal generator (i.e. a finite output impedance of the generator voltage source):

1. Nonparametric methods for LTI systems

1.k Nonlinear systems operating in feedback (cont'd)

Example interaction between generator/actuator and the dynamical system



43/316

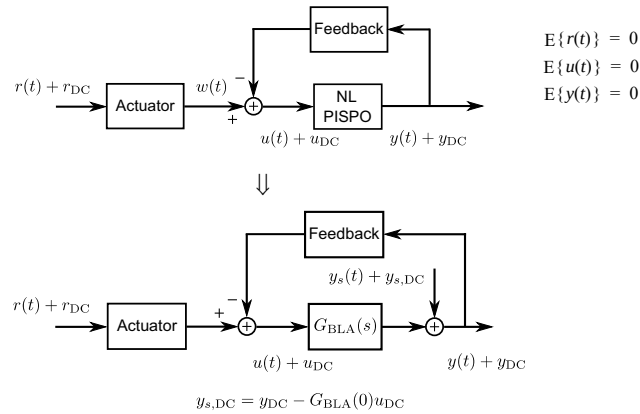
The electrical circuit is converted into the block diagram by working out the equations:

$$\begin{cases} e = Zi + v \\ v = z_{NL}\{i\} \end{cases} \Rightarrow i = \frac{1}{Z}e - \frac{1}{Z}z_{NL}\{i\} \quad (2.16)$$

The BLA is now defined as the ratio of cross spectra, respectively of the reference and the output S_{yr} and of the reference and the input S_{ur} .

1. Nonparametric methods for LTI systems

1.k Nonlinear systems operating in feedback (cont'd)



Definition best linear approximation (BLA):

$$G_{BLA}(j\omega) = \frac{S_{yr}(j\omega)}{S_{ur}(j\omega)} = \frac{F\{E\{y(t)r(t-\tau)\}\}}{F\{E\{u(t)r(t-\tau)\}\}}$$

where the expected value is taken w.r.t. the random realisation of $r(t)$

44/316

It will be clear further on that above definition is equivalent with the open loop definition of the BLA. Also here, in the frequency domain, the definition uses the cross- and auto-power spectra (in steady-state):

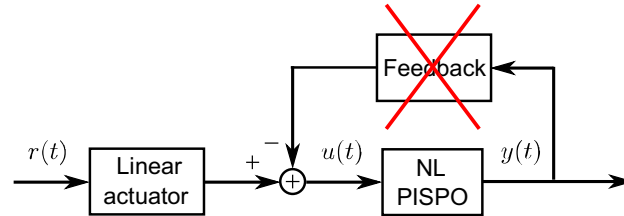
$$G_{BLA}(j\omega_k) = \frac{\mathbb{E}\{Y(k)\overline{R(k)}\}}{\mathbb{E}\{U(k)\overline{R(k)}\}} \quad (2.17)$$

This definition is also valid for the open loop case with actuator dynamics.

1. Nonparametric methods for LTI systems

1.k Nonlinear systems operating in feedback (cont'd)

Nonlinear system operating in open loop



$$G_{\text{BLA}}(j\omega) = \frac{S_{yr}(j\omega)}{S_{ur}(j\omega)} = \frac{S_{yu}(j\omega)\overline{G_{\text{act}}^{-1}}(j\omega)}{S_{uu}(j\omega)\overline{G_{\text{act}}^{-1}}(j\omega)} = \frac{S_{yu}(j\omega)}{S_{uu}(j\omega)}$$

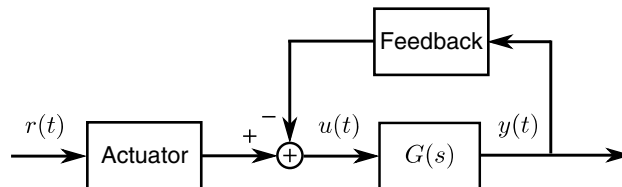
46/316

If the feedback is nonlinear we have

1. Nonparametric methods for LTI systems

1.k Nonlinear systems operating in feedback (cont'd)

Linear system operating in a nonlinear feedback loop



$$G_{\text{BLA}}(j\omega) = \frac{S_{yr}(j\omega)}{S_{ur}(j\omega)} = \frac{G(j\omega)S_{ur}(j\omega)}{S_{ur}(j\omega)} = G(j\omega)$$

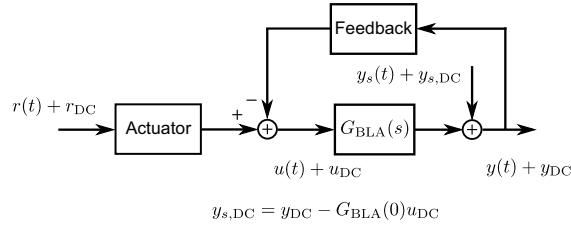
$$Y_s(k) = Y(k) - G(j\omega_k)U(k) = 0$$

47/316

If the plant is nonlinear we have (where r_{DC} , u_{DC} and y_{DC} are constants defining the setpoint of the setup)

1. Nonparametric methods for LTI systems

1.k Nonlinear systems operating in feedback (cont'd)



General class of random excitations $r(t)$ with given *joint pdf* and *power spectrum*

$y_s(t)$:

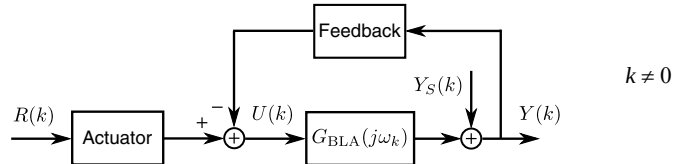
- zero mean
- uncorrelated with – but *not* independent of – $r(t)$
- not normally distributed

48/316

Expressed in the frequency domain

1. Nonparametric methods for LTI systems

1.k Nonlinear systems operating in feedback (cont'd)



Class of *Gaussian* excitation signals $r(t)$ with given Riemann equivalent *power spectrum*

$Y_S(k)$:

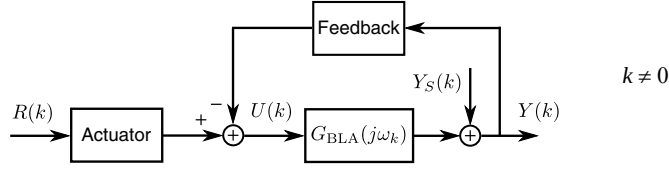
- $E\{Y_S(k)\bar{R}(k)\} = 0 \Rightarrow Y_S(k)$ uncorrelated with but *not* independent of $R(k)$
- $E\{Y_S(k)\} = 0$
- $Y_S(k)$ is asymptotically uncorrelated over k (cumulant mixing over k)
- asymptotically normally distributed
- $E\{|Y_S(k)|^2\}$ is a smooth function of the frequency

Note: expectations are taken over the different random realisations of the reference signal

49/316

1. Nonparametric methods for LTI systems

1.k Nonlinear systems operating in feedback (cont'd)



Best linear approximation open loop system from reference to input-output simultaneously

$$\begin{aligned} U(k) &= G_{ru}(j\omega_k)R(k) + \tilde{U}_S(k) \\ Y(k) &= G_{ry}(j\omega_k)R(k) + \tilde{Y}_S(k) \end{aligned} \quad \text{where } \tilde{U}_S(k), \tilde{Y}_S(k): \text{ uncorrelated with } R(k)$$

Best linear approximation from input to output

$$G_{BLA}(j\omega_k) = \frac{E\{Y(k)\bar{R}(k)\}}{E\{U(k)\bar{R}(k)\}} = \frac{G_{ry}(j\omega_k)}{G_{ru}(j\omega_k)}$$

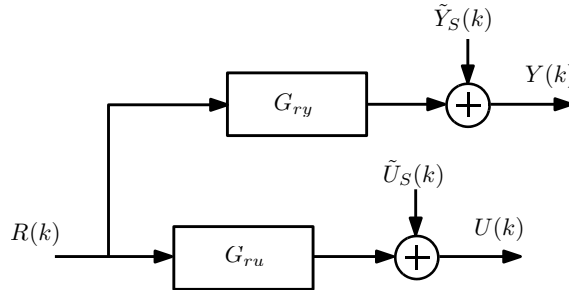
Relationship observed input-output distortions $\tilde{U}_S(k)$, $\tilde{Y}_S(k)$ and output residual $Y_S(k)$

$$Y_S(k) = Y(k) - G_{BLA}(j\omega_k)U(k) = \tilde{Y}_S(k) - G_{BLA}(j\omega_k)\tilde{U}_S(k)$$

50/316

See also [3].

In this case, the spectra $U(k)$ and $Y(k)$ are both considered as output spectra. To make this explicit, we can rewrite the feedback schematic as a Single input 2 output system:



The expression $G_{BLA}(j\omega_k) = \frac{G_{ry}(j\omega_k)}{G_{ru}(j\omega_k)}$ is obtained as follows:

1. The BLA G_{ru} from the reference R to the input U is

$$G_{ru} = G_{\text{actuator}} \frac{1}{1 + G_{BLA}F}$$

2. The BLA G_{ry} from R to the output Y is

$$G_{ry} = G_{\text{actuator}} \frac{G_{BLA}}{1 + G_{BLA}F}$$

and dividing both gives G_{BLA} .

Note also that

$$G_{ry} = \frac{S_{yr}}{S_{rr}}, \quad G_{uy} = \frac{S_{ur}}{S_{rr}} \quad \Rightarrow \quad G_{BLA} = \frac{G_{ry}}{G_{ru}} = \frac{S_{yr}}{S_{ur}} \quad (2.18)$$

(because $\tilde{U}_S, \tilde{Y}_S$ are uncorrelated with $R(k)$) as defined earlier on.

The spectra $\tilde{U}_S$ and $\tilde{Y}_S$ are the nonlinear distortions due to the *complete* feedback mechanism, and observed on the spectra $U(k)$ and $Y(k)$. In contrast, Y_S is the nonlinear distortion due to the system G only. They are related as follows:

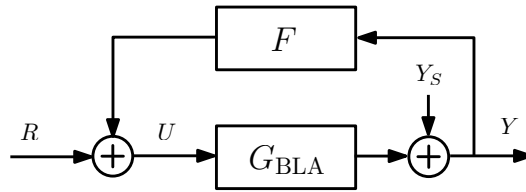
$$Y_s = Y - G_{\text{BLA}}U = G_{ry}R + \tilde{Y}_s - G_{\text{BLA}}(G_{ru}R + \tilde{U}_s) \quad (2.19)$$

$$= \tilde{Y}_s - G_{\text{BLA}}\tilde{U}_s \quad (2.20)$$

because: $G_{ry} = G_{\text{BLA}}G_{ru}$.

2.5.1 1st order estimation of a nonlinearity in closed loop

From equation (2.20), and by using the concepts from the ‘fast’ method (Section 2.4.3), we can obtain an estimate of the nonlinear distortions created by the plant, from a single experiment.



$$\hat{G}_{\text{BLA}}(j\omega_{k_e}) = \frac{G_{ry}(j\omega_{k_e})}{G_{ru}(j\omega_{k_e})} \quad (2.21)$$

$$\hat{Y}_S = Y_{\text{ne}} - \hat{G}_{\text{BLA,interp}}U_{\text{ne}} \quad (2.22)$$

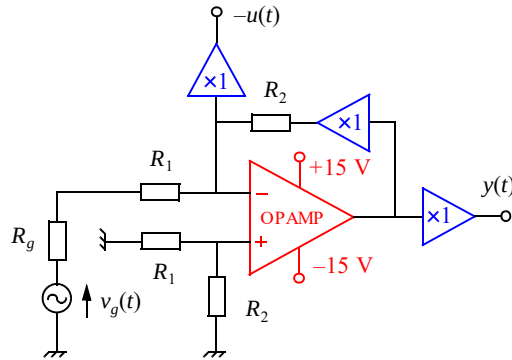
Here, $G_{\text{BLA,interp}}$ is an interpolation of $\hat{G}_{\text{BLA}}$ from the excited frequencies to the unexcited frequencies. Y_{ne} and U_{ne} are the output and input spectra respectively at the not excited frequencies.

If the reference signal is not available, the estimate $\hat{G}_{\text{BLA}} = \frac{Y}{U}$ could be used. However, note that this estimate is biased, precisely due to the nonlinear distortions Y_S in the feedback loop (which have a similar effect on the estimate as the process noise, see Exercise 2.1, p. 3).

The estimation of the nonlinear distortions Y_s in a feedback loop is illustrated on an example with an OPAMP, below.

1. Nonparametric methods for LTI systems

1.1 Measurement example: the open loop gain of an operational amplifier



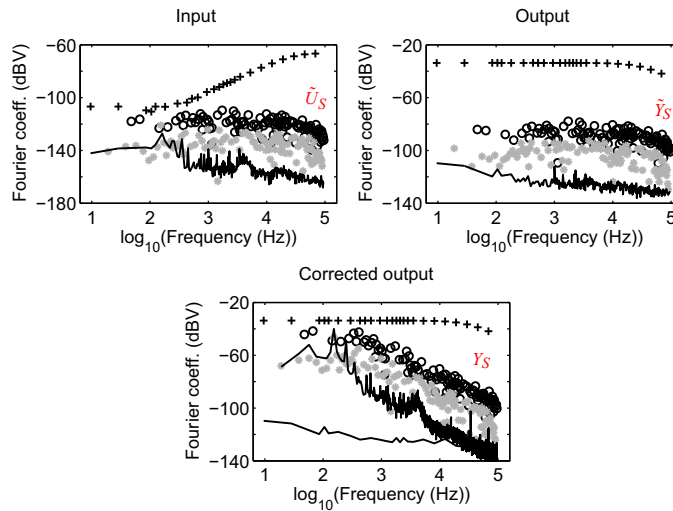
Excitation $v_g(t)$:

- odd random phase multisine with random harmonic grid
- logarithmic frequency distribution between 10 Hz and 100 kHz
- $P = 5$ periods, $M = 25$ experiments, $N = 64 \times 1024$ samples per period,
- $f_s = 625$ kHz

51/316

1. Nonparametric methods for LTI systems

1.1 Measurement example: the open loop gain of an operational amplifier (cont'd)



52/316

As is seen, the ‘directly observed’ nonlinear distortions $\tilde{Y}_S$ in the output spectrum are very small. However, the nonlinear distortions are clearly present in the input, as $\tilde{U}_S$. After correction, the nonlinear distortions Y_S are actually much higher than what would have been expected from $\tilde{Y}_S$ alone. This is an illustration of a feedback loop suppressing nonlinear distortions in the output.

2.6 Detection and quantification of time-varying systems

Refer to course on *Advanced Measurement and Data Driven Modelling*.

Parametric models of LTI systems

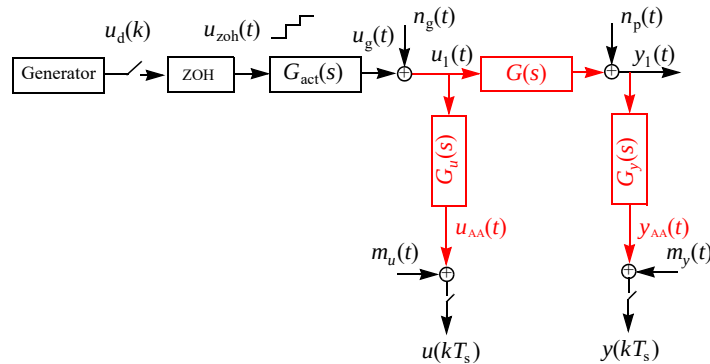
3.1 Band-limited vs Zero-Order-Hold setups

For the *Band-limited* (BL) setup, the system is modelled from its actual input to output, where the input and output signals are filtered (anti-alias filters G_u and G_y) such as to avoid any alias.

2. Parametric models of LTI systems

2.a Band-limited versus zero-order-hold measurement set up (cont'd)

Band-limited (BL)



- errors-in-variables problem
- actuator is not part of the **continuous-time model**
- perfect BL set up
 - $G_u(j\omega) = G_y(j\omega)$ for $|f| < f_s/2 \Rightarrow$ relative calibration
 - $G_u(j\omega) = G_y(j\omega) = 0$ for $|f| \geq f_s/2$

94/316

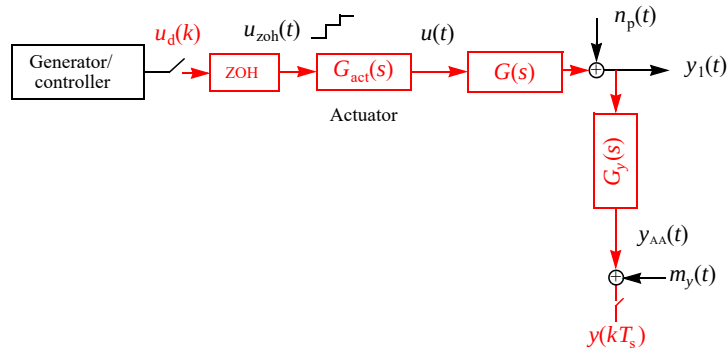
The BL setup is useful when we are interested in a *continuous-time model* of $G(s)$ (i.e. independently of the actuator dynamics), e.g. for physical interpretations of the measured system.

For the *Zero-order-hold* (ZOH) setup, the system is modelled from the *discrete-time signal* in the actuator (known exactly) to the *discretised* (noisy) output signal. No anti-alias filters are present. The step-like signals introduced by the zero-order-hold generate high-frequency components (beyond the Nyquist frequency), which cause alias. This is taken into account in the expression for $G_{ZOH}(z^{-1})$.

2. Parametric models of LTI systems

2.a Band-limited versus zero-order-hold measurement set up

Zero-order-hold (ZOH)



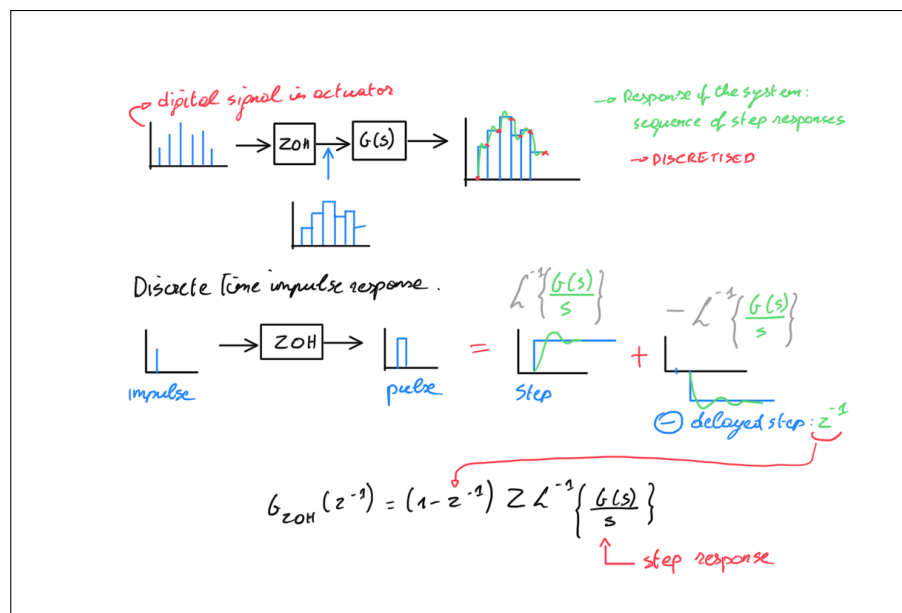
- input known exactly $\Rightarrow$ output error problem
- actuator is part of the **discrete-time model**
- perfect ZOH set up
 - $G_{\text{act}}(j\omega) = 1$, $G_y(j\omega) = 1$ from DC to $\infty \Rightarrow$ absolute calibration
 - $G_{\text{ZOH}}(z^{-1}) = (1 - z^{-1})Z\{L^{-1}\{G(s)/s\}\}$

93/316

where Z represents the Z-transform of the sampled signals.

The ZOH setup is relevant when the dynamic behaviour of the system needs to be determined, in the context of a given instrumental setup (DAC/ADC, actuator dynamics, ...). This is for example the case for *control purposes*, i.e. when the system is controlled by a discrete-time controller. Note that the resulting model does depend on the instrumental setup.

The expression for G_{ZOH} is obtained as the Z-transform of sampled step responses $\frac{G(s)}{s}$ of the plant. This is because the ZOH transforms a discrete time impulse to a continuous square pulse of width one sample (which is the difference between a step and a delayed step). Graphically explained as follows.



Summary of the BL vs ZOH setups:

BL	ZOH
Two channel measurement	Single channel measurement
Relative calibration	Absolute calibration
Instrument bandwidth $\approx f_s/2$	Instrument bandwidth $\approx kf_s, k \gg 1$
Anti-alias filters required	Anti-alias filters not allowed
Continuous-time models	Discrete-time models
Physical modelling (only the DUT)	Control applications (whole measurement chain)

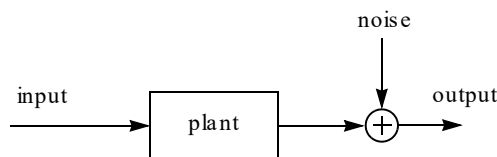
3.1.1 Band-limited and ZOH noise models

If the noise needs to be described parametrically, it is important to formulate the model structure of the noise. We will consider that the noise is the response of an LTI system to white noise (i.e. filtered white noise). The noise properties are then completely described by this filter.

2. Parametric models of LTI systems

2.d Band-limited versus zero-order-hold noise models

Problem statement



Noise mostly continuous-time:

- thermal noise
- $1/f$ noise in semi-conductor devices
- ...

Goal:

- write noise at output as filtered white noise

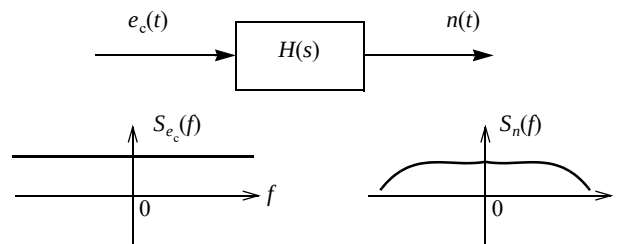
Reason:

- white noise source needed to prove asymptotic properties estimators
- nonparametric noise model $N \times N$ covariance matrix with Toeplitz structure

105/316

2. Parametric models of LTI systems

2.d Band-limited versus zero-order-hold noise models (cont'd)



First trial

- $e_c(t)$ = white CT-noise

Problem

- either infinite variance or zero power spectrum

106/316

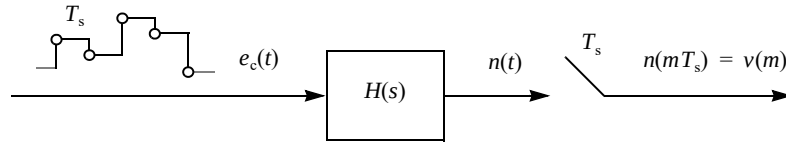
Continuous-time white noise has a constant power spectrum (up to ∞). Therefore, it has an infinite power, and is not handy to work with. So, some tricks are needed.

We will consider the noise at the sample instants only. This is *discrete-time* noise, for which there is no problem to be white (i.e. to have a constant power spectrum). Still, the question remains whether we can determine a relationship with some continuous-time noise that satisfies the discrete-time (sampled) model used.

If **discrete time models** are at hand, we determine what a continuous-time equivalent would be of discrete time white noise. As an intersample behaviour, we make the assumption of ZOH.

2. Parametric models of LTI systems

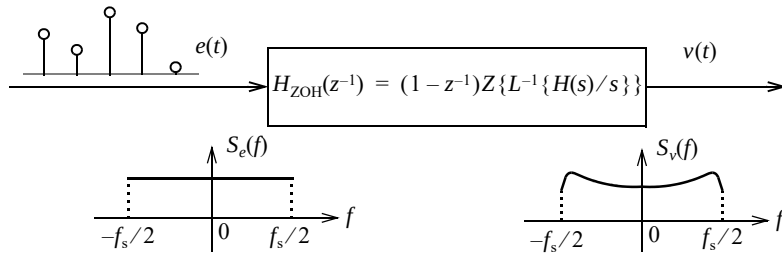
2.d Band-limited versus zero-order-hold noise models (cont'd)



Assumptions ZOH noise model

- $e_c(t)$ is piecewise constant where $e(m) = e_c(mT_s)$ is white DT-noise
- no anti-alias filter

At sampling instances



Note: if $e_c(t)$ is a Wiener stochastic process (integrated CT white noise) $\Rightarrow$ exact DT model

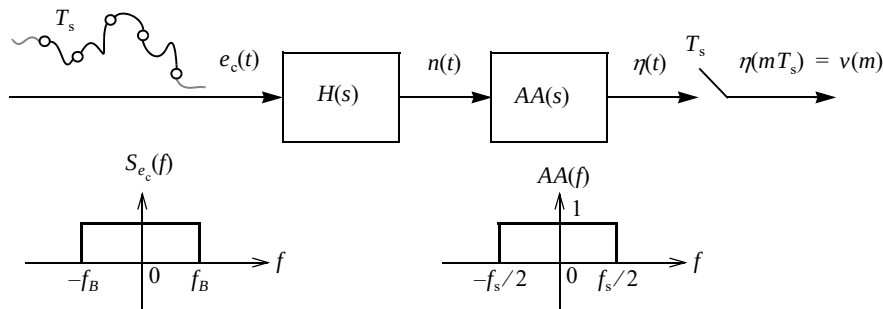
107/316

In this case, $H_{ZOH}(z^{-1})$ is the discretised version of $H(s)$ via the step-invariant transformation. To understand the relationship between both, if p_c is a pole of $H(s)$, then $p_d = e^{p_c T_s}$ is a pole of $H_{ZOH}(z^{-1})$.

If **continuous time models** are at hand, we can start from continuous-time noise, which is white *in the frequency band of interest* (i.e. at least in $[-f_s, f_s]$), and then work with a band-limited measurement setup. In this case, it's as if the noise was generated by a continuous-time white noise, restricted in frequency span by the anti-alias filter (AA), and resulting in a discrete time white noise sequence $v(m)$.

2. Parametric models of LTI systems

2.d Band-limited versus zero-order-hold noise models (cont'd)



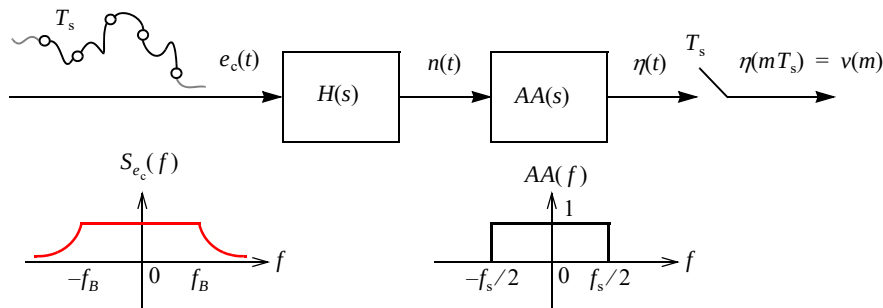
Assumptions BL noise model

- $e_c(t)$ = band-limited white CT-noise (Åström, 1970)
- $e_c(t)$ is Gaussian distributed
- band-limited measurement setup
- $f_B \geq f_s/2$

108/316

2. Parametric models of LTI systems

2.d Band-limited versus zero-order-hold noise models (cont'd)



Relaxed assumption on the driving CT noise source $e_c(t)$

- $S_{e_c}(f)$ = is constant in the band $|f| < f_B$
- $S_{e_c}(f) = O(f^{-(1+\delta)})$ for $|f| \geq f_B$ and $\delta > 0$

where $O(f^{-(1+\delta)})$ is the weakest decay such that $\text{var}(e_c(t)) = \int_{-\infty}^{+\infty} S_{e_c}(f) df < \infty$

110/316

The relaxed assumption is such that the driving continuous-time noise source e_c is not exactly band-limited but still has finite power.

3.2 Lumped versus distributed systems

Lumped systems are described by Ordinary Differential Equations (ODE), while distributed systems are described by Partial Differential Equations (PDEs).

Examples of

Lumped systems: Low-frequency ($\ll 100$ MHz) electronic circuits, mass-spring-damper systems, stiff robotic systems, ...

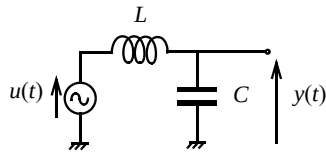
Distributed systems: Mass- and heat-transfer systems (e.g. diffusion processes), vibrating structures (bridges, buildings), High-frequent ($\gg 100$ MHz) electronic circuits, ...

Lumped systems, like an LC circuit have a transfer function given by a rational transfer function in s , of finite order.

2. Parametric models of LTI systems

2.e Lumped versus distributed LTI systems

Lumped system



$$LC \frac{d^2 y(t)}{dt^2} + y(t) = u(t)$$

$$\text{initial conditions: } y(0) = 0, \left. \frac{dy(t)}{dt} \right|_{t=0} = 0$$

transfer function

$$G(s) = \frac{Y(s)}{U(s)} = \frac{1}{1 + LCs^2}$$

poles

$$s = \pm j / \sqrt{LC}$$

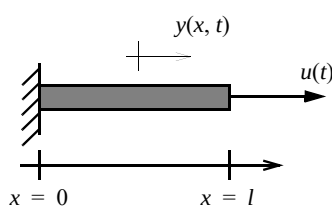
113/316

An example of a *distributed* system is a vibrating beam (longitudinal vibrations in the figure). Note that, in this case, the order of the transfer function reaches ∞ , that is, it has an infinite number of resonances (known as *modes*). However, the number of active resonances in a limited frequency range is finite, such that the transfer function can be approximated by a rational form in that band.

2. Parametric models of LTI systems

2.e Lumped versus distributed LTI systems (cont'd)

Distributed system



$$\frac{\partial^2 y(x,t)}{\partial t^2} = \frac{E}{\rho} \frac{\partial^2 y(x,t)}{\partial x^2}$$

initial conditions: $y(x, 0) = 0, \left. \frac{\partial y(x,t)}{\partial t} \right|_{t=0} = 0$

boundary conditions: $y(0, t) = 0, \left. \frac{\partial y(x,t)}{\partial x} \right|_{x=l} = \frac{u(t)}{E}$

transfer function

$$G(s) = \frac{Y(l,s)}{U(s)} = \frac{l \tanh(\tau s)}{E \tau s} = \frac{l}{E} \sum_{k=0}^{\infty} \frac{2}{(\tau s)^2 + (\pi(2k+1)/2)^2} \text{ with } \tau = \sqrt{\frac{\rho l^2}{E}}$$

poles

$$s = \pm(2k+1) \frac{\pi}{2\tau} j, k = 1, 2, \dots$$

Note

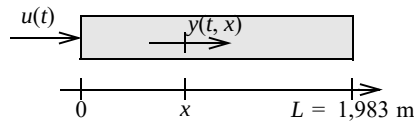
$$|2 / ((\tau s)^2 + (\pi(2k+1)/2)^2)|_{s=j\omega}$$

has limited active frequency range $\Rightarrow$ rational approximation in given frequency band

114/316

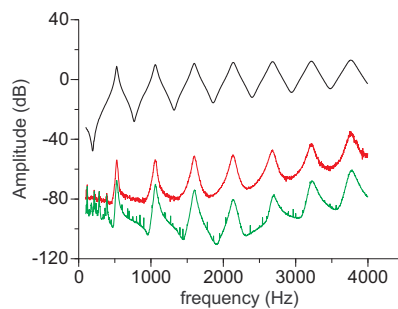
5. Estimation parametric plant model with estimated nonparametric noise model

5.c Measurement example: identification of Young's modulus



plexiglass beam
1 cm $\times$ 2 cm $\times$ 1.983 m

Force-to-acceleration



$M = 25, P = 10$

— measured FRF
— total variance
— noise variance

221/316

2. Parametric models of LTI systems

2.f Lumped systems

- Continuous-time (CT)
General

$$\sum_{n=0}^{n_a} a_n y^{(n)}(t) = \sum_{m=0}^{n_b} b_m u^{(m)}(t) \Rightarrow G(s, \theta) = \frac{B(s, \theta)}{A(s, \theta)} = \frac{\sum_{r=0}^{n_b} b_r s^r}{\sum_{r=0}^{n_a} a_r s^r}$$

Sometimes (diffusion phenomena such as heat or mass transfer)

$$\sum_{n=0}^{n_a} a_n \frac{d^{n/2} y(t)}{dt^{n/2}} = \sum_{m=0}^{n_b} b_m \frac{d^{m/2} u(t)}{dt^{m/2}} \Rightarrow G(\sqrt{s}, \theta) = \frac{B(\sqrt{s}, \theta)}{A(\sqrt{s}, \theta)} = \frac{\sum_{m=0}^{n_b} b_m s^{m/2}}{\sum_{n=0}^{n_a} a_n s^{n/2}}$$

- Discrete-time (DT)

$$\sum_{n=0}^{n_a} a_n y(t-n) = \sum_{m=0}^{n_b} b_m u(t-m) \Rightarrow G(z^{-1}, \theta) = \frac{B(z^{-1}, \theta)}{A(z^{-1}, \theta)} = \frac{\sum_{r=0}^{n_b} b_r z^{-r}}{\sum_{r=0}^{n_a} a_r z^{-r}}$$

- Transient behaviour
 - differential and difference equations: exponential decay
 - fractional differential equation: $O(t^{-3/2})$ decay

115/316

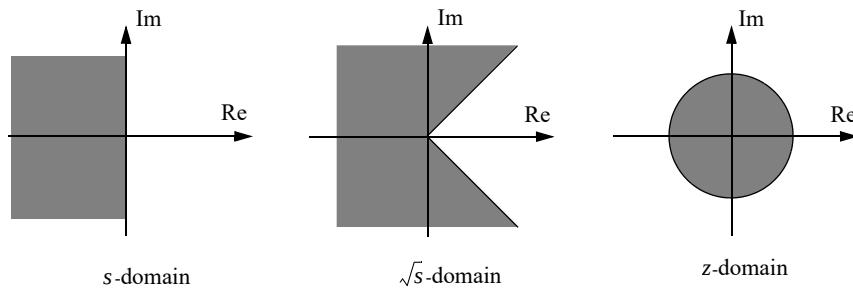
For diffusion systems, described approximately by rational forms in $\sqrt{s}$, the stability region is given by the allowed locations of the poles in $\sqrt{s}$ such that the resulting pole in s is in the left half plane.

3.3 A generalised model structure and frequency variable Ω

2. Parametric models of LTI systems

2.f Lumped systems (cont'd)

Summary



$$G(\Omega, \theta) = \frac{B(\Omega, \theta)}{A(\Omega, \theta)} = \frac{\sum_{r=0}^{n_b} b_r \Omega^r}{\sum_{r=0}^{n_a} a_r \Omega^r} \text{ with } \Omega = \begin{cases} s & \text{continuous-time} \\ \sqrt{s} & \text{diffusion} \\ z^{-1} & \text{discrete-time} \end{cases}$$

116/316

2. Parametric models of LTI systems

2.g Relation between input/output DFT spectra - plant model only

• Input/output DFT spectra

$$U(k) = \frac{1}{\sqrt{N}} \sum_{t=0}^{N-1} u(tT_s) z_k^{-t}, \quad Y(k) = \frac{1}{\sqrt{N}} \sum_{t=0}^{N-1} y(tT_s) z_k^{-t}$$

$$\text{with } z_k = e^{j2\pi k/N}$$

• Periodic signals

$$Y(k) = G(\Omega_k, \theta) U(k)$$

if

- steady state response
- integer number of periods are observed

117/316

2. Parametric models of LTI systems

2.g Relation between input/output DFT spectra - plant model only (cont'd)

• Arbitrary signals

$$Y(k) = G(\Omega_k, \theta) U(k) + T_G(\Omega_k, \theta) + \delta(\Omega_k)$$

with

$$G(\Omega, \theta) = \frac{B(\Omega, \theta)}{A(\Omega, \theta)} = \frac{\sum_{r=0}^{n_b} b_r \Omega^r}{\sum_{r=0}^{n_a} a_r \Omega^r}, \quad T_G(\Omega, \theta) = \frac{I_G(\Omega, \theta)}{A(\Omega, \theta)} = \frac{\sum_{r=0}^{n_i} i_{g_r} \Omega^r}{\sum_{r=0}^{n_a} a_r \Omega^r}$$

where

i_{g_r} = function of the difference between the initial and final conditions

$$n_{i_g} = \max(n_a, n_b) - 1 \text{ for } \Omega = z^{-1} \text{ and } n_{i_g} = \max(n_a, n_b) \text{ for } \Omega = s, \sqrt{s}$$

$$|U(k)| = O(N^0), \quad |T_G(\Omega_k, \theta)| = O(N^{-1/2})$$

$$\delta(\Omega_k) = \begin{cases} 0 & \Omega = z^{-1} \\ \text{residual alias error } O(N^{-1/2}) & \Omega = s, \sqrt{s} \end{cases}$$

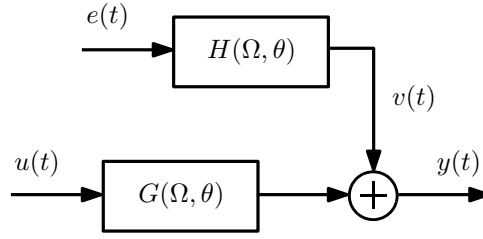
Notes:

- $T_G(\Omega, \theta) = \delta(\Omega_k) = 0$ when the initial conditions equal the final conditions
- reduce $\delta(\Omega_k)$ for $\Omega = s, \sqrt{s}$ by choice $n_{i_g} \geq \max(n_a, n_b)$

118/316

3.3.1 Parametrisation of rational forms

Considering the following configuration:



we denote

$$G(\Omega, \theta) = \frac{B(\Omega, \theta)}{A(\Omega, \theta)} \quad T_G(\Omega_k, \theta) = \frac{I_G(\Omega_k, \theta)}{A(\Omega_k, \theta)} \quad (3.1)$$

$$H(\Omega, \theta) = \frac{C(\Omega, \theta)}{D(\Omega, \theta)} \quad T_H(\Omega_k, \theta) = \frac{I_H(\Omega_k, \theta)}{D(\Omega_k, \theta)} \quad (3.2)$$

and we have that

$$Y(k) = G(\Omega_k, \theta)U(k) + T_g(\Omega_k, \theta) + H(\Omega_k, \theta)E(k) + T_H(\Omega_k, \theta) \quad (3.3)$$

Typical special cases are given by

ARX :	$C = 1, D = A$
ARMAX :	$D = A, n_{i_g} \geq \max(n_a, n_b, n_c) - 1$
ARMA :	$G = 0, T_G = 0$
BJ :	$D \neq A$

with

- AR: autoregressive
- MA: moving average
- X: exogenous input
- BJ: Box-Jenkins

2. Parametric models of LTI systems

2.m Plant and noise model structures (cont'd)

• DT domain

$$y(t) = G(q, \theta)u(t) + H(q, \theta)e(t)$$

with $q = z^{-1}$ the backward shift operator: $qx(t) = x(t-1)$

and where

$$G(q)x(t) = \left(\sum_{r=0}^{\infty} g(r)q^r\right)x(t) = \sum_{r=0}^{\infty} g(r)x(t-r)$$

• CT domain

$$y(t) = G(p, \theta)u(t) + H(p, \theta)e(t)$$

with $p = \frac{d}{dt}$ the derivative operator: $px(t) = \frac{dx(t)}{dt}$

and where

$$G(p)x(t) = \int_0^{\infty} g(\tau)x(t-\tau)d\tau$$

129/316

2. Parametric models of LTI systems

2.n Parametrizations LTI systems

Plant model

• Rational form

$$G(\Omega, \theta) = \frac{B(\Omega, \theta)}{A(\Omega, \theta)} = \frac{\sum_{r=0}^{n_b} b_r \Omega^r}{\sum_{r=0}^{n_a} a_r \Omega^r} \text{ with } \theta^T = [a_0 a_1 \dots a_{n_a} b_0 b_1 \dots b_{n_b}]$$

• Partial fraction expansion

$$G(\Omega, \theta) = \sum_{\substack{r=-p \\ r \neq 0}}^p \frac{L_r}{\Omega - \lambda_r} + \sum_{r=1}^q \frac{S_r}{\Omega - \sigma_r} + W(\Omega, w) \text{ for } \Omega = s, \sqrt{s}$$

$$G(z^{-1}, \theta) = \sum_{\substack{r=-p \\ r \neq 0}}^p \frac{L_r z^{-1}}{1 - \lambda_r z^{-1}} + \sum_{r=1}^q \frac{S_r z^{-1}}{1 - \sigma_r z^{-1}} + W(z^{-1}, w)$$

with

$$\theta^T = [\sigma_1 \dots \sigma_q \text{Re}(\lambda_1) \text{Im}(\lambda_1) \dots \text{Re}(\lambda_p) \text{Im}(\lambda_p) \dots \\ S_1 \dots S_q \text{Re}(L_1) \text{Im}(L_1) \dots \text{Re}(L_p) \text{Im}(L_p) w_0 \dots w_{n_w}]$$

130/316

2. Parametric models of LTI systems

2.n Parametrizations LTI systems (cont'd)

Plant model (cont'd)

- State space representation for proper transfer functions ($n_b \leq n_a$)

$$G(s, \theta) = C(sI_{n_a} - A)^{-1}B + D$$

$$G(z^{-1}, \theta) = z^{-1}C(I_{n_a} - z^{-1}A)^{-1}B + D$$

with

$$\theta^T = [\text{vec}^T(A) \ B^T \ C \ D]$$

- Pole/zero representation

$$G(\Omega, \theta) = K \frac{\prod_{r=1}^{n_b} (\Omega - \zeta_r)}{\prod_{r=1}^{n_a} (\Omega - \lambda_r)}$$

disadvantage: ill conditioned for multiple poles/zeros

- Systems with time delay

$$G(\Omega, \theta) = e^{-\tau s} \frac{B(\Omega, \theta)}{A(\Omega, \theta)}$$

$$G(z^{-1}, \theta) = z^{-\tau/T_s} \frac{B(z^{-1}, \theta)}{A(z^{-1}, \theta)}$$

131/316

- Bond-graphs
- Differential Algebraic Equations (DAE)
- State-space

Not necessarily input-output data.

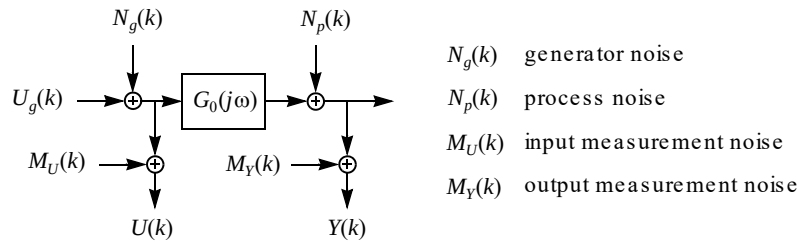
Parametric transfer function identification of LTI systems

4.1 Maximum likelihood estimation, with known non-parametric noise model

If the noise model is known exactly, it is often possible to formulate the maximum likelihood estimator. This is the setup.

4. Estimation parametric plant model with known noise model

4.a Frequency domain data



Noisy input/output observations

$$\begin{cases} Y(k) = Y_0(k) + N_Y(k) \\ U(k) = U_0(k) + N_U(k) \end{cases} \text{ with known } \begin{cases} \sigma_Y^2(k) = \text{var}(N_Y(k)) = E\{|N_Y(k)|^2\} \\ \sigma_U^2(k) = \text{var}(N_U(k)) = E\{|N_U(k)|^2\} \\ \sigma_{YU}^2(k) = \text{covar}(N_Y(k), N_U(k)) = E\{N_Y(k)\overline{N_U(k)}\} \end{cases}$$

- periodic signals

$$U_0(k) = U_g(k)$$

- arbitrary signals

$$U_0(k) = U_g(k) + N_g(k)$$

187/316

If an estimate (and the uncertainty) of the FRF is available,

$$G(j\omega_k) = G_0(j\omega_k) + N_G(k), \quad \text{with known } \sigma_G^2(k) = \text{var}(N_G(k)), \quad (4.1)$$

this can be considered as a special case of noisy input/output observations, with

$$Y(k) = G(j\omega_k) \quad N_Y(k) = N_G(k) \quad (4.2)$$

$$U(k) = 1 \quad N_U(k) = 0. \quad (4.3)$$

The maximum likelihood estimator starts from the probability density function of the measurements, and maximises this w.r.t. the parameters. In this case, the parameters consist of the model parameters θ , and (more surprisingly) the true values of the input/output spectra, denoted

$$Z_p(k) = \begin{bmatrix} Y_p(k) \\ U_p(k) \end{bmatrix}. \quad (4.4)$$

4. Estimation parametric plant model with known noise model

4.b Maximum likelihood estimator

Define

$$Z(k) = [Y(k), U(k)]^T \text{ and } N_Z(k) = [N_Y(k) \ N_U(k)]^T$$

Assumption

$N_Z(k)$ has zero mean and is independent (over k), circular complex normally distributed

Gaussian likelihood function

$$\begin{aligned} f_Z(Z|\theta, Z_p) &= \prod_{k=1}^F f_{Z(k)}(Z(k)|\theta, Z_p(k)) \\ &= \frac{1}{\pi^{2F} \prod_{k=1}^F \det(C_Z(k))} \exp(-\sum_{k=1}^F (Z(k) - Z_p(k))^H C_Z^+(k) (Z(k) - Z_p(k))) \end{aligned}$$

with

$$Z_p(k) = \begin{bmatrix} Y_p(k) \\ U_p(k) \end{bmatrix} \text{ and } C_Z(k) = \text{Cov}(N_Z(k)) = \begin{bmatrix} \sigma_Y^2(k) & \sigma_{YU}^2(k) \\ \bar{\sigma}_{YU}^2(k) & \sigma_U^2(k) \end{bmatrix}$$

and where $C_Z(k)$ can be singular (e.g. feedback with process noise and no measurement noise)

Note:

- $C_Z^+(k)$ is the Moore-Penrose pseudo-inverse of $C_Z(k)$

189/316

4. Estimation parametric plant model with known noise model

4.b Maximum likelihood estimator (cont'd)

• Periodic signals

Gaussian ML cost function

$$V_{ML}(\theta, Z_p, Z) = \sum_{k=1}^F (Z(k) - Z_p(k))^H C_Z^+(k) (Z(k) - Z_p(k)) + \sum_{k=1}^F \lambda_k [1, -G(\Omega_k, \theta)] Z_p(k)$$

where $G(\Omega, \theta)$ is parametrized according to one of the plant models in chapter 2

Eliminate $Z_p = [Z_p(1), Z_p(2), \dots, Z_p(F)]^T, \lambda \Rightarrow$

$$V_{ML}(\theta, Z) = \sum_{k=1}^F \frac{|e(\Omega_k, \theta, Z(k))|^2}{\sigma_e^2(\Omega_k, \theta)}$$

where

$$e(\Omega_k, \theta, Z(k)) = Y(k) - G(\Omega_k, \theta) U(k)$$

$$\sigma_e^2(\Omega_k, \theta) = \sigma_Y^2(k) + |G(\Omega_k, \theta)|^2 \sigma_U^2(k) - 2\text{Re}(\bar{G}(\Omega_k, \theta) \sigma_{YU}^2(k))$$

• Arbitrary signals

Same cost function with

$$e(\Omega_k, \theta, Z(k)) = Y(k) - G(\Omega_k, \theta) U(k) - T_G(\Omega_k, \theta)$$

190/316

Eliminating Z_p and λ is not straightforward. Proof in [2, Section 9.11] and its appendix.

Note that V_{ML} results in a very common cost function, consisting of the weighted (σ_e^2) sum of absolute squared errors $|e|^2$.

4. Estimation parametric plant model with known noise model

4.b Maximum likelihood estimator (cont'd)

• Discussion

$$V_{ML}(\theta, Z) = \sum_{k=1}^F \frac{|e(\Omega_k, \theta, Z(k))|^2}{\sigma_e^2(\Omega_k, \theta)}$$

with

$$\begin{cases} e(\Omega_k, \theta, Z(k)) = Y(k) - G(\Omega_k, \theta)U(k) - T_G(\Omega_k, \theta) \\ \sigma_e^2(\Omega_k, \theta) = \sigma_Y^2(k) + |G(\Omega_k, \theta)|^2 \sigma_U^2(k) - 2\text{Re}(\bar{G}(\Omega_k, \theta) \sigma_{YU}^2(k)) \end{cases}$$

- suppresses frequency bands with poor signal-to-noise ratio

- relative importance input noise w.r.t. output noise (uncorrelated case $\sigma_{YU}^2 = 0$)

$$\frac{|G(\Omega_k, \theta)|^2 \sigma_U^2(k)}{\sigma_Y^2(k)}$$

- significance correlation between input/output errors

$$\rho(k) = \frac{-2\text{Re}(\sigma_{YU}^2(k) \bar{G}(\Omega_k, \theta))}{\sigma_Y^2(k) + |G(\Omega_k, \theta)|^2 \sigma_U^2(k)} \text{ with } \begin{cases} \rho(k) < 0 \Rightarrow \text{decrease } \sigma_e^2(\Omega_k, \theta) \\ \rho(k) > 0 \Rightarrow \text{increase } \sigma_e^2(\Omega_k, \theta) \end{cases}$$

191/316

4. Estimation parametric plant model with known noise model

4.b Maximum likelihood estimator (cont'd)

• Properties

- Consistency

$$V_F(\theta) = E \left\{ \frac{1}{F} V_{ML}(\theta, Z) \right\} = E \left\{ \frac{1}{F} \sum_{k=1}^F \frac{|e(\Omega_k, \theta, Z_0(k))|^2}{\sigma_e^2(\Omega_k, \theta)} \right\} + 1$$

expected value cost function $V_F(\theta)$ is minimal in $\theta = \theta_0 \Rightarrow$ consistent

- Convergence rate

$$\hat{\theta}(Z) = \theta_0 + \delta_\theta(Z) + b_\theta(Z)$$

$$\delta_\theta(Z) = -V_F^{-1}(\theta_0) V_F'^T(\theta_0, Z)$$

with

$$\delta_\theta(Z) = O_p(F^{-1/2}) \text{ and } E\{\delta_\theta(Z)\} = 0: \text{dominating stochastic error}$$

$$b_\theta(Z) = O_p(F^{-1}): \text{bias error}$$

- Asymptotic normality

$$\sqrt{F}(\hat{\theta}(Z) - \theta_0) \in \text{AsN}(0, \text{Cov}(\sqrt{F} \delta_\theta(Z)))$$

with

$$\text{Cov}(\sqrt{F} \delta_\theta(Z)) = V_F^{-1}(\theta_0) Q_F(\theta_0) V_F^{-1}(\theta_0)$$

$$Q_F(\theta_0) = F E\{V_F'^T(\theta_0, Z) V_F'(Z, \theta_0)\}$$

193/316

The consistency is determined by checking that the expected value of the cost function is minimal in the true parameter values. It is demonstrated that the noisy part of the cost function has the expected value 1 (independently of θ), and that

the noiseless part (i.e. using Z_0) is 0 (and thus minimal) in the true parameter values θ_0 .

That the estimate is biased with $b_\theta(Z) = \mathcal{O}_p(F^{-1})$ is due to (residual) transient/leakage errors [2, Appendix 9.E.3]. This bias is typically negligible over the variance.

4.1.1 Approximate maximum likelihood as initial estimates

The problem of the maximum likelihood cost function is that it is nonlinear in the parameters and, thus, non-convex. A *linear least squares* convex relaxation is proposed.

4. Estimation parametric plant model with known noise model

4.e Starting values

- Basic idea for any parametrization

Two possibilities

- rational transfer function + linear least squares + back transformation
- state space + subspace algorithm + back transformation

- Linear least squares

$$V_{LS}(\theta, Z) = \sum_{k=1}^F |e(\Omega_k, \theta, Z(k))|^2 = \sum_{k=1}^F |A(\Omega_k, \theta)Y(k) - B(\Omega_k, \theta)U(k) - I_G(\Omega_k, \theta)|^2$$

Properties

- inconsistent

$$E\{V_{LS}(\theta, Z)/F\} = \frac{1}{F} \sum_{k=1}^F |e(\Omega_k, \theta, Z_0(k))|^2 + \frac{1}{F} \sum_{k=1}^F \sigma_e^2(\Omega_k, \theta)$$

is not minimal in θ_0

- overemphasizes high frequency errors
- estimate is constraint dependent since $V_{LS}(\lambda\theta, Z) = \lambda^2 V_{LS}(\theta, Z)$
- a well chosen weighting can be added $\Rightarrow$ WLS
- $\tilde{\theta}(Z_0) = \arg \min E\{V_{LS}(\theta, Z)\}$ depends on the noise level

198/316

Estimating a rational form $G = B/A$ naturally leads to a *non-identifiability* issue. Namely, scaling all the parameters (numerator and denominator) by the same value λ will result in the same model, viz.:

$$G(\lambda\theta) = G(\theta), \quad \text{for } \lambda \neq 0 \quad (4.5)$$

Therefore, a constraint should be applied (see further). However, if the cost function *does* depend on this scaling, then it is shown in [2] that the estimate will depend on the chosen constraint on the parameters. This is the case for the Linear least squares estimate.

4. Estimation parametric plant model with known noise model

4.e Starting values (cont'd)

- Iterative quadratic maximum likelihood (IQML)

Iterative algorithm

$$V_{\text{IQML}}(\theta^{(i+1)}, Z) = \sum_{k=1}^F \frac{|e(\Omega_k, \theta^{(i+1)}, Z(k))|^2}{\sigma_e^2(\Omega_k, \theta^{(i)})}$$

started from (W)LS solution

- Other methods

- generalized total least squares (consistent + amplification high frequency errors)
- bootstrapped total least squares (consistent + close to ML + not self starting)
- subspace (consistent, frequency weighting not clear)

199/316

Here is an overview of the different estimators to generate starting values for the (nonlinear) Maximum Likelihood estimator.

LLS (Linear least squares)

The cost function is V_{LS} . It is a quadratic form in the parameters, resulting in a convex problem with a single (global) minimizer, solved by using the pseudo-inverse. It does not take the noise properties into account properly, resulting in an inconsistent estimate. Linear constraints on the parameters are required (e.g. $a_i = 1$ or $b_i = 1$). The result depends on the constraint used, which gives problems when the chosen constraint is inappropriate (e.g. constraint used is $a_0 = 1$, but actually $a_0 \approx 0$). In many cases, this estimator *overemphasizes high frequency errors*.

IWLS (Iterative Weighted Least Squares)

The Iterative Weighted Least Squares (IWLS) has the cost function

$$V_{\text{IWLS}}^{(i)}(\theta^{(i)}, Z) = \sum_{k=1}^F \frac{|e(\Omega_k, \theta^{(i)}, Z(k))|^2}{|A(\Omega_k, \theta^{(i-1)})|^2} \quad (4.6)$$

This *compensates* the *overemphasized high frequency errors* of the LLS. Just like the LLS, this estimate has the advantage that it does not need the noise information. However, both the LLS and the IWLS are not consistent (i.e. they do not use the noise information appropriately). The IWLS is also known as the Sanathanan and Koerner technique, as mentioned in [2, Section 9.8.3]. The IWLS also needs a (linear) constraint.

IQML (Iterative Quadratic Maximum Likelihood)

Instead of using the denominator of the transfer function (i.e. IWLS), the noise variance of the equation error is used:

$$V_{\text{IQML}}(\theta^{(i)}) = \sum_{k=1}^F \frac{|e(\Omega_k, \theta^{(i)})|^2}{\sigma_e^{2r}(\Omega_k, \theta^{(i-1)})} \quad (4.7)$$

The IQML gives a much improved frequency weighting. We set $r \in [0, 1]$, where typically $r = 1$ is chosen to have the closest resemblance with the Maximum Likelihood, but a smaller r results in a more convex problem ($r = 0$ is equivalent with the LS estimator).

TLS (Total Least Squares)

The cost function is V_{LS} , but with constraint $\|\theta\|_2^2 = 1$:

$$V_{\text{TLS}}(\theta) = V_{\text{LS}}(\theta) \quad \text{s.t. } \|\theta\|_2^2 = 1 \quad (4.8)$$

Its minimizer can be shown [2, Lemma 9.23] to be equivalent to

$$\hat{\theta}_{\text{TLS}} = \underset{\theta}{\operatorname{argmin}} \frac{\|e(\theta)\|_2^2}{\|\theta\|_2^2} \quad (4.9)$$

This choice of constraint is more robust to ill-conditioned problems (i.e. it also works if one of the coefficients $a_i, b_i = 0$). However, it also has the problem of LLS, that the HF errors are overemphasized. Also, that the noise weight is not appropriate results in an inconsistent estimate.

The inconsistency can be demonstrated by computing the expected value of the cost function, and observing that its minimum depends on the noise properties. Denote $J\theta = e(\theta)$ (with J the regression matrix, given that $e(\theta)$ is linear in θ). Then,

$$V_{\text{TLS}}(\theta) = \frac{\|J\theta\|_2^2}{\|\theta\|_2^2} \quad (4.10)$$

Splitting J in its noiseless and noisy parts:

$$J = J_o + \Delta J, \quad (4.11)$$

the expected value of the noisy part of V_{TLS} gives

$$\mathbb{E}\{v_{\text{TLS}}(\theta)\} = \frac{\theta^H \mathbb{E}\{\Delta J^H \Delta J\} \theta}{\|\theta\|_2^2} = \frac{\theta^H C_J \theta}{\|\theta\|_2^2} \quad (4.12)$$

which remains θ -dependent.

To **compute** $\hat{\theta}_{\text{TLS}}$, use the Singular Value Decomposition (SVD) of J :

$$J = U_J \Sigma_J V_J^H \quad (4.13)$$

Where the columns of U_J and V_J are called the left and right singular vectors respectively. Also, U_J and V_J are unitary: $U_J^H U_J = I$ and $V_J^H V_J = I$, and Σ_J is a diagonal matrix of singular values (ordered from largest to smallest). Then, we

have that $\hat{\theta}_{\text{TLS}}$ is the right singular vector corresponding to the smallest singular value.

$$\hat{\theta}_{\text{TLS}} = V_{J,[:,\text{end}]} \quad (4.14)$$

Proof: as an exercise.

An extension to the TLS can be made to make it consistent. This estimator is called the Generalised Total Least Squares (GTLS) [2, Section 9.10.3].

GTLS (Generalized Total Least Squares)

The GTLS extends the TLS with appropriate row weighting $C_J^{-\frac{1}{2}}$, to make it consistent, while still remaining convex (because quadratic in the parameters). Specifically,

$$V_{\text{GTLS}}(\theta) = \|J\theta\|_2^2 \quad \text{subject to} \quad \|C_J^{-\frac{1}{2}}\theta\|_2^2 = 1 \quad (4.15)$$

with J the linear regression matrix of the problem (s.t. $J\theta = e(\theta)$), $C_J = \mathbb{E}\{\Delta J^H \Delta J\}$ the column covariance of J , and ΔJ the noisy part of J . By following the reasoning for (4.12) on (4.15), it is clear that the expected value of the GTLS estimator is independent of the noise, and minimal in the true parameter vector.

To solve (4.15), rewrite it as

$$V_{\text{GTLS}}(\theta) = \|JC_J^{-\frac{1}{2}}\tilde{\theta}\|_2^2 \quad \text{subject to} \quad \|\tilde{\theta}\|_2^2 = 1 \quad (4.16)$$

with $\tilde{\theta} = C_J^{-\frac{1}{2}}\theta$. If $C_J^{-\frac{1}{2}}$ is singular (which is often the case, e.g. with noiseless regressors for the transients or noiseless in- or output signals), then the problem can be solved by using the GSVD (Generalised Singular Value Decomposition) of the matrix pair $(J, C_J^{-\frac{1}{2}})$, as follows. Denote this GSVD as:

$$J = UCX^T \quad (4.17)$$

$$C_J^{-\frac{1}{2}} = V SX^T \quad (4.18)$$

with U and V unitary, C and S diagonal (elements $\{c_i\}, \{s_i\}$), and X a regular matrix. Choose

$$\hat{\theta}_{\text{GTLS}} = (X^T)_{[:,i]}^{-1} \frac{1}{s_i}, \quad \text{s.t. } i = \underset{i}{\text{argmin}} \frac{c_i}{s_i}. \quad (4.19)$$

Then, $J\hat{\theta} = U_i \frac{c_i}{s_i}$, which is minimal in 2-norm by choosing i such that the smallest generalised singular value is selected $\min \frac{c_i}{s_i}$. Thus, $\hat{\theta}_{\text{GTLS}}$ is the generalised singular vector corresponding to the smallest generalised singular value. Also, note that $\|C_J^{-\frac{1}{2}}\hat{\theta}\|_2^2 = \|V_i\|_2^2 = 1$, such that the norm constraint is satisfied.

The GTLS is developed in further details in [2, Section 9.10.3]. The GTLS is consistent, but it still suffers from higher weights at the high frequencies, thus losing in efficiency. This problem is (partly) solved by the Iterative Quadratic Maximum Likelihood estimator (IQML).

BTLS (Bootstrapped Total Least Squares)

This is the GTLS extended with the same improved frequency weighting as the IQML in (4.7). Thus, the BTLS is iterative, with the following extension w.r.t. GTLS:

$$J \rightarrow WJ \quad \text{with } W = \text{diag } \sigma_e^{-2r} \left(\Omega_k, \theta^{(i-1)} \right) \quad (4.20)$$

The BTLS is developed in further details in [2, Section 9.12.3].

4.1.2 Imposing real parameter values

A real system will be described by a transfer function, as in page 37, with *real* coefficients. However, the optimisation of the (approximate) maximum likelihood cost function minimizes the magnitude of *complex* errors, such that there is no guarantee that the estimated parameters will be real. In general, the estimates will be complex.

To impose that the estimated parameters are real, first express the equation error as:

$$e = J\theta = (\Re J + j\Im J)(\Re\theta + j\Im\theta) \quad (4.21)$$

$$\Leftrightarrow \begin{bmatrix} \Re e \\ \Im e \end{bmatrix} = \begin{bmatrix} \Re J & -\Im J \\ \Im J & \Re J \end{bmatrix} \begin{bmatrix} \Re\theta \\ \Im\theta \end{bmatrix} \quad (4.22)$$

and set $\Im\theta = 0$. This gives:

$$\begin{bmatrix} \Re e \\ \Im e \end{bmatrix} = \begin{bmatrix} \Re J \\ \Im J \end{bmatrix} \Re\theta \triangleq J_{\text{re}}\theta \quad (4.23)$$

Thus, to impose real estimated parameters, a new regression matrix J_{re} needs to be constructed, which simply concatenates the real and imaginary parts of the complex regression matrix J .

For the GTLS estimate, **the column covariance matrix** of J_{re} must be computed:

$$C_{J_{\text{re}}} = \mathbb{E} \{ \Delta J_{\text{re}}^T \Delta J_{\text{re}} \} \quad (4.24)$$

To this end, first split ΔJ into its real and imaginary parts:

$$\Delta J = \Re \Delta J + j\Im \Delta J \quad (4.25)$$

Then, C_J is computed as

$$C_J = \mathbb{E} \{ \Delta J^H \Delta J \} \quad (4.26)$$

$$= \mathbb{E} \{ \Re \Delta J^T \Re \Delta J + \Im \Delta J^T \Im \Delta J \} \quad \leftarrow \text{Real part} \quad (4.27)$$

$$+ j\mathbb{E} \{ \Re \Delta J^T \Im \Delta J - \Im \Delta J^T \Re \Delta J \} \quad \leftarrow \text{Imaginary part} \quad (4.28)$$

By definition, $J_{\text{re}} = \begin{bmatrix} \Re J \\ \Im J \end{bmatrix}$, giving

$$C_{J_{\text{re}}} = \mathbb{E} \{ \Delta J_{\text{re}}^T \Delta J_{\text{re}} \} \quad (4.29)$$

$$= \mathbb{E} \{ \Re \Delta J^T \Re \Delta J \} + \mathbb{E} \{ \Im \Delta J^T \Im \Delta J \} \quad (4.30)$$

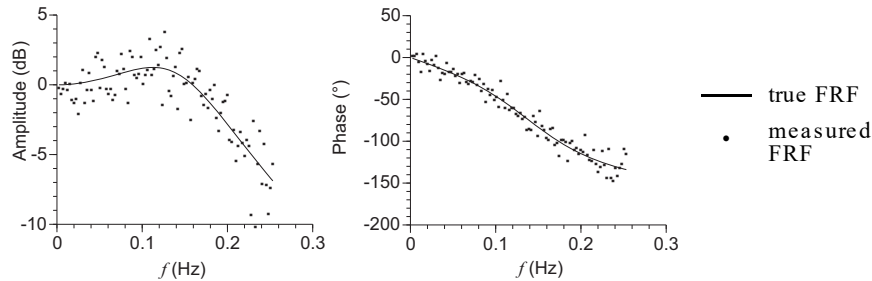
$$= \Re C_J \quad (4.31)$$

So, to compute $C_{J_{re}}$, simply take the real part of C_J .

4. Estimation parametric plant model with known noise model

4.f Simulation example: 2nd order continuous-time system

$$G(s, \theta_0) = 1/(1 + s + s^2)$$



Simulation parameters

- $U_0(k) = 1$
- $N_U(k), N_Y(k) \in N^c(0, 0.02)$
- $F = 100$ in the band $[0.1, 10]/(2\pi)^2$ Hz

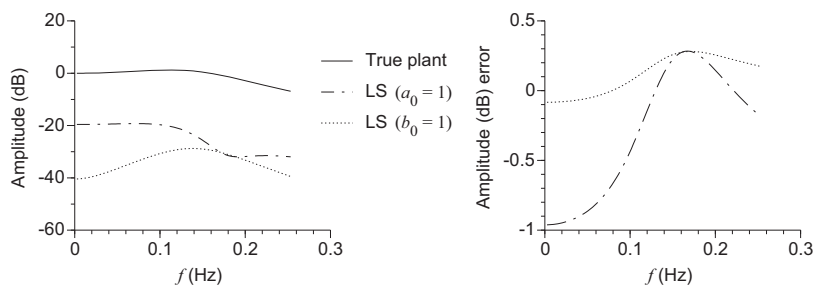
200/316

The dependence on the choice of the constraint is illustrated below.

4. Estimation parametric plant model with known noise model

4.f Simulation example: 2nd order continuous-time system (cont'd)

Linear least squares



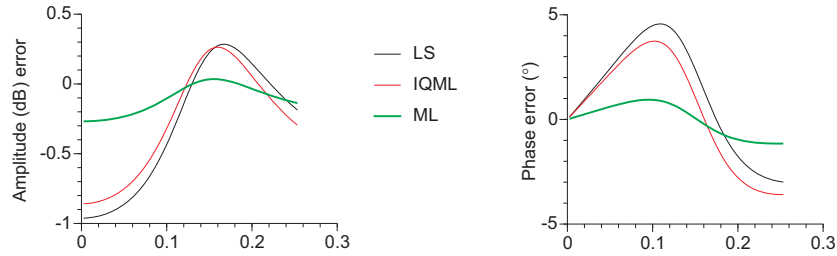
Observe the influence of the constraint on the LS estimate:

- $a_0 = 1 \Rightarrow$ underbiased
- $b_0 = 1 \Rightarrow$ overbiased

201/316

4. Estimation parametric plant model with known noise model

4.f Simulation example: 2nd order continuous-time system (cont'd)



Notes:

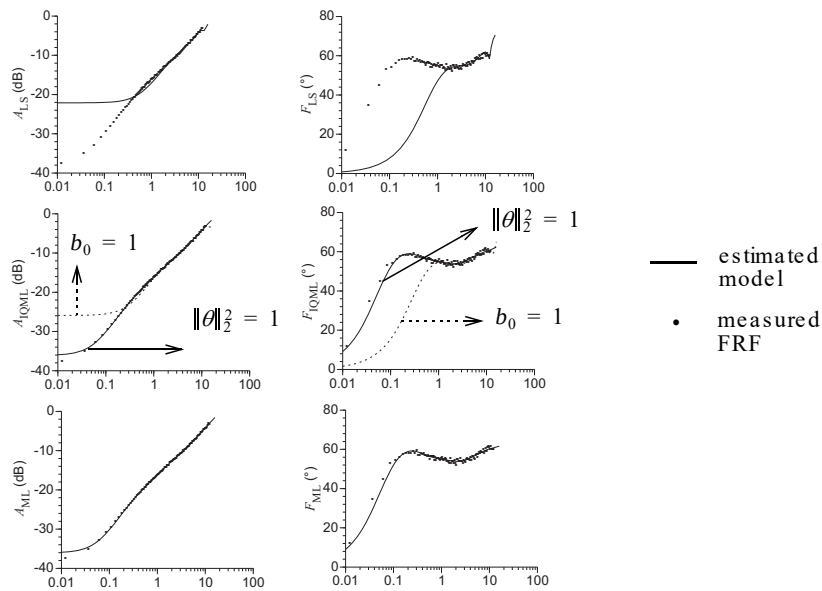
- LS uses no noise information
- IQML, and ML use noise information

203/316

4. Estimation parametric plant model with known noise model

4.g Measurement example: q-axis impedance 3.4 MW synchronous machine

$$n_b = 4, n_a = 3$$

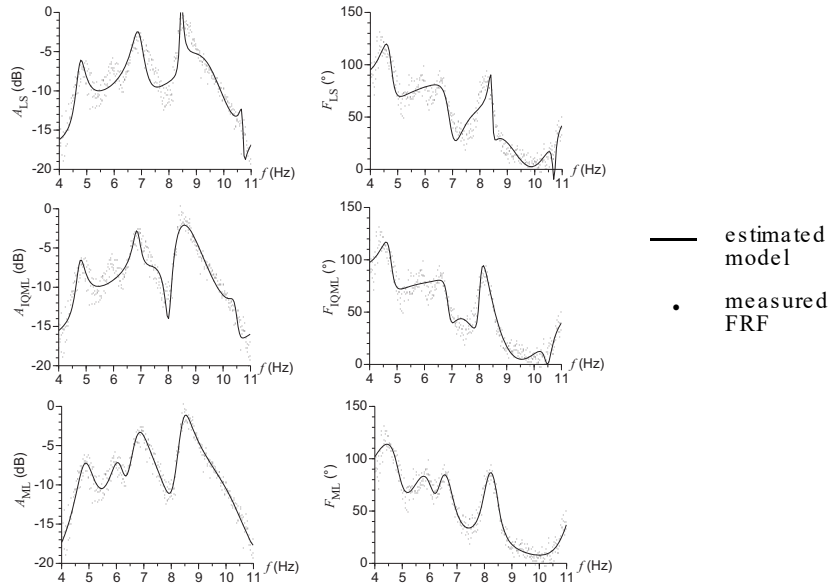


204/316

4. Estimation parametric plant model with known noise model

4.h Measurement example: flight flutter data

$$n_b = 11, n_a = 10$$



205/316

4.1.3 Computing the maximum likelihood estimate

The maximum likelihood estimator minimizes a nonlinear least squares cost function. See Section 8.4 in the lecture notes on *Measurement and Data Driven Modeling* for an elaboration on the minimisation of a nonlinear least squares cost function.

4. Estimation parametric plant model with known noise model

4.c Calculation ML estimates

• Newton-Gauss

Basic algorithm

$$\text{Re}(J^{(i)H} J^{(i)}) \Delta \theta^{(i+1)} = -\text{Re}(J^{(i)H} \mathcal{E}^{(i)})$$

where

$$\mathcal{E}_{[k]}^{(i)} = \frac{e(\Omega_k, \theta^{(i)}, Z(k))}{\sigma_e(\Omega_k, \theta^{(i)})} \text{ and } J_{[k,r]}^{(i)} = \frac{\partial \mathcal{E}_{[k]}^{(i)}}{\partial \theta_{[r]}^{(i)}}$$

Numerical stable implementation

$$J_{\text{re}}^{(i)} \Delta \theta^{(i+1)} = -\mathcal{E}_{\text{re}}^{(i)} \Rightarrow \Delta \theta^{(i+1)} = -(J_{\text{re}}^{(i)})^+ \mathcal{E}_{\text{re}}^{(i)} = -V^{(i)}(\Sigma^{(i)})^+ U^{(i)T} \mathcal{E}_{\text{re}}^{(i)}$$

where

$$X_{\text{re}} = \begin{bmatrix} \text{Re}(X) \\ \text{Im}(X) \end{bmatrix} \text{ and } J_{\text{re}}^{(i)} = U^{(i)} \Sigma^{(i)} V^{(i)T}$$

Notes:

- continuous-time, scale angular frequencies by median frequency set
 $\omega_{\text{med}} = \text{med}\{\omega_1, \omega_2, \dots, \omega_F\} \Rightarrow b_r s^r = (b_r \omega_{\text{med}}^r)(s/\omega_{\text{med}})^r$
- leave all parameters free and impose in each iteration step the parameter constraint, for example, $\|\theta\|_2^2 = 1$
- normalize the columns of $J_{\text{re}}^{(i)}$ by their 2-norm

195/316

The Newton-Gauss algorithm is an approximation of the Newton-Raphson method, where the terms with the second derivatives of the regression matrix are

neglected.

4. Estimation parametric plant model with known noise model

4.c Calculation ML estimates (cont'd)

• Levenberg-Marquardt

Basic algorithm

$$(J_{\text{re}}^{(i)T} J_{\text{re}}^{(i)} + \lambda^2 I) \Delta \theta^{(i+1)} = -J_{\text{re}}^{(i)T} \mathcal{E}_{\text{re}}^{(i)}$$

with

$$\lambda_{\text{start}} = \sigma_{\max}(J_{\text{re}}) / 100$$

cost function decreases: $\lambda \rightarrow 0,4\lambda$

cost function increases: $\lambda \rightarrow 10\lambda$

Numerical stable implementation for the constraint $\|\theta\|_2 = 1$

$$\Delta \theta^{(i+1)} = -V^{(i)} A^{(i)} U^{(i)T} \mathcal{E}_{\text{re}}^{(i)}$$

with

$$J_{\text{re}} = U \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_{n_\theta-1}, 0) V^T$$

$$A = \text{diag}\left(\frac{\sigma_1}{\sigma_1^2 + \lambda^2}, \frac{\sigma_2}{\sigma_2^2 + \lambda^2}, \dots, \frac{\sigma_{n_\theta-1}}{\sigma_{n_\theta-1}^2 + \lambda^2}, 0\right)$$

196/316

σ_i is the i th singular value of J_{re} .

One step of the Levenberg Marquardt algorithm

$$\begin{aligned} (J^T J + \lambda^2 I) \Delta \theta &= -J^T \mathcal{E} \quad \text{with } J = \frac{d\mathcal{E}}{d\theta} \\ \text{SVD of } J, \text{ with last singular value set to } 0 \quad & \text{(take into account the rank deficiency)} \\ J &= U \Sigma_0 V^T \\ & \text{last element} = 0 \\ (U \Sigma_0 \cancel{U^T} \cancel{U} \Sigma_0 V^T + \lambda^2 I) \Delta \theta &= -U \Sigma_0 U^T \mathcal{E} \\ & \quad \lambda^2 V V^T \\ \underbrace{V(\Sigma_0^2 + \lambda^2 I) V^T}_{\text{diagonal matrix}} \Delta \theta &= -U \Sigma_0 U^T \mathcal{E} \\ \Rightarrow \Delta \theta &= -V(\Sigma_0^2 + \lambda^2 I)^{-1} \Sigma_0 U^T \mathcal{E} \end{aligned}$$

4.1.4 Computing the covariance of the maximum likelihood estimates

4. Estimation parametric plant model with known noise model

4.d Calculation covariance matrix ML estimates

Fix a coefficient, for example, of the largest denominator power of Ω , and remove the corresponding column in the Jacobian matrix

$$\text{Cov}(\hat{\theta}) \approx (2\text{Re}(J^H J))^{-1} = (2J_{\text{re}}^T J_{\text{re}})^{-1} = \frac{1}{2}(V\Sigma^{-1})(V\Sigma^{-1})^T$$

with

$$J_{[k,r]} = \frac{\partial \varepsilon_{[k]}}{\partial \hat{\theta}_{[r]}}$$

$$\varepsilon_{[k]} = \frac{e(\Omega_{k^*}, \theta, Z(k))}{\sigma_e(\Omega_{k^*}, \theta)}$$

$$J_{\text{re}} = \begin{bmatrix} \text{Re}(J) \\ \text{Im}(J) \end{bmatrix} = U\Sigma V^T$$

Note: perform the calculations on the frequency normalized parameters for s - and $\sqrt{s}$ -domains

197/316

To understand the factor 2 in $\text{cov}(\hat{\theta}) = (2\Re(J^H J))^{-1}$, note that, for

$$\mathbb{E} \{ \varepsilon \varepsilon^H \} = I \quad \varepsilon_{\text{re}} = \begin{bmatrix} \Re \varepsilon \\ \Im \varepsilon \end{bmatrix} \quad (4.32)$$

and ε circular complex, we have that

$$\mathbb{E} \{ \Re \varepsilon \Re \varepsilon^H \} = \mathbb{E} \{ \Im \varepsilon \Im \varepsilon^H \} = \frac{1}{2} I. \quad (4.33)$$

4.1.5 Covariance of the estimated poles and zeroes

By using $\text{Cov}(\theta)$, the covariance of the estimated parameters, the covariance of the estimated poles can be calculated as well. Consider a small variation in the parameters, $\delta\theta$, resulting in a variation in the poles δp^* . Via a first order approximation, we have:

$$\delta p = \frac{dp}{d\theta} \delta\theta \quad (4.34)$$

Then, the covariance of the poles is computed as the covariance of δp :

$$\text{Cov}(p) = \mathbb{E} \{ \delta p \delta p^H \} = \mathbb{E} \left\{ \frac{dp}{d\theta} \delta\theta \delta\theta^H \frac{dp}{d\theta}^H \right\} = \frac{dp}{d\theta} \text{Cov}(\theta) \frac{dp}{d\theta}^H \quad (4.35)$$

The derivative $\frac{dp}{d\theta}$ is computed as follows. Consider the polynomial (representing the numerator or denominator of the transfer function):

$$f(\theta, p) = \theta_2 p^2 + \theta_1 p + \theta_0. \quad (4.36)$$

The roots (poles or zeroes) are the solutions of $f(\theta, p^*) = 0$. The derivative of the roots are obtained via the partial derivatives of the implicit function¹

$$\left. \frac{dp}{d\theta} \right|_{p=p^*} = - \frac{\frac{\partial f}{\partial \theta}}{\frac{\partial f}{\partial p}} = - \frac{[p^{*2} \quad p^* \quad 1]}{2\theta_2 p^* + \theta_1} \quad (4.38)$$

which can easily be extended to higher order polynomials. This is also described in Appendix A of [1].

4.1.6 Model structure and complexity selection, and model validation

In general, the true system complexity can be very high. However, it is usually desirable to derive a model that fits the data well with a low complexity. A good trade-off between bias and variance is searched.

Indications of whether a model complexity is sufficient:

- comparison of the estimated parametric transfer function with the measured (non-parametric) FRF: the difference should have the same order of magnitude as the standard deviation of the FRF
- value of the cost function, compared to expected value
- level of the residuals, compared to their uncertainty
- other statistical tests (typically on the residuals): whiteness test of (weighted residuals)

4. Estimation parametric plant model with known noise model

4.i Model validation

- Comparison with measured FRF

Compare $G(\Omega_k, \hat{\theta})$ to the nonparametric LPM estimate (see Chapter 3)

$\hat{G}_{\text{fast}}(\Omega_k), \hat{G}_{\text{robust}}(\Omega_k)$	for periodic signals
$\hat{G}_{\text{arb}}(\Omega_k)$	for arbitrary signals

taking into account the uncertainty $\sigma_G^2(k)$ of the LPM estimate

- Analysis cost function

No modelling errors

$E\{V_{\text{ML}}(\hat{\theta}_{\text{ML}}(Z), Z)\} \approx V_{\text{noise}}$ $\text{var}(V_{\text{ML}}(\hat{\theta}_{\text{ML}}(Z), Z)) \approx V_{\text{noise}}$	with	$V_{\text{noise}} = F - n_\theta/2$ $n_\theta = \text{number of free parameters}$
---	------	--

Test

$$V_{\text{ML}}(\hat{\theta}_{\text{ML}}(Z), Z) > V_{\text{noise}} + 2\sqrt{V_{\text{noise}}} \Rightarrow \text{modelling errors}$$

$$V_{\text{ML}}(\hat{\theta}_{\text{ML}}(Z), Z) \in [V_{\text{noise}} - 2\sqrt{V_{\text{noise}}}, V_{\text{noise}} + 2\sqrt{V_{\text{noise}}}] \Rightarrow \text{no modelling errors}$$

$$V_{\text{ML}}(\hat{\theta}_{\text{ML}}(Z), Z) < V_{\text{noise}} - 2\sqrt{V_{\text{noise}}} \Rightarrow \text{wrong noise variances?}$$

206/316

¹Non-rigorous derivation of the derivative of an implicit function: set the differential of f to 0:

$$df = \frac{\partial f}{\partial \theta} d\theta + \frac{\partial f}{\partial p} dp = 0 \quad (4.37)$$

and solve for $\frac{dp}{d\theta}$.

4. Estimation parametric plant model with known noise model

4.i Model validation (cont'd)

- Whiteness test residuals measured minus modelled FRF

Sample correlation residuals

$$\hat{R}_{\delta\delta}(m) = \frac{1}{F-m} \sum_{k=1}^{F-m} \delta(\Omega_k, \hat{\theta}) \overline{\delta(\Omega_{k+m}, \hat{\theta})} \quad \text{with } \delta(\Omega_k, \hat{\theta}) = \frac{\hat{G}(\Omega_k) - G(\Omega_k, \hat{\theta})}{\sigma_{\hat{G}}(\Omega_k)}$$

No unmodelled dynamics

$$- 95\% \text{ confidence bound } \sqrt{3} \text{std}(\hat{R}_{\delta\delta}(m)) = \sqrt{3/(F-m)}$$

- $\hat{R}_{\delta\delta}(m)$ is asymptotically ($F \rightarrow \infty$) a Dirac function

$$\begin{cases} \hat{R}_{\delta\delta}(m) = O(F^{-1/2}) & \text{for } m \neq 0 \\ \hat{R}_{\delta\delta}(0) \geq 1 \end{cases}$$

- $\hat{R}_{\delta\delta}(0) = 1$ and noise (co-)variances $\Rightarrow$ no nonlinear distortions

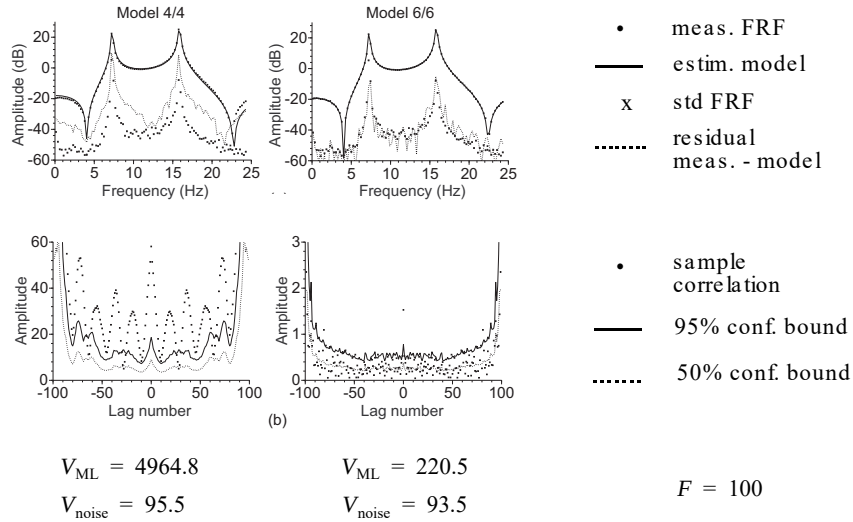
- $\hat{R}_{\delta\delta}(0) > 1$ and noise (co-)variances $\Rightarrow$ nonlinear distortions

- (co-)variances noise + stochastic nonlinear distortions $\Rightarrow \hat{R}_{\delta\delta}(0) = 1$

207/316

4. Estimation parametric plant model with known noise model

4.j Measurement example: flexible robot arm



Observation: $\hat{R}_{\delta\delta}(0) > 1$ and $\sigma_G^2 = \text{var}(N_G) \Rightarrow$ nonlinear distortions

208/316

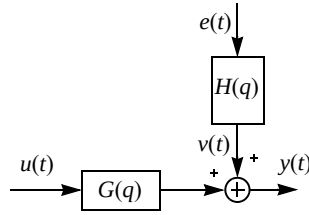
4.2 Simultaneous noise and plant model estimation

In this case, the purpose is to estimate a parametric model of both the plant G and the noise filter H . As before, the input $e(t)$ to the noise filter is *not* observed. A popular approach is to use the “prediction error framework” (PEM), in the discrete time case.

7. Estimation parametric plant and/or noise models

7.a Classical time domain prediction error framework

• Discrete-time plant and noise model



Model

$$y(t) = G(q)u(t) + H(q)e(t)$$

with a monic noise model

$$H(z^{-1}) = \frac{1 + c_1 z^{-1} + \dots}{1 + d_1 z^{-1} + \dots}$$

Notes:

- $u(t)$ and $y(t)$ are known exactly and $e(t)$ is not observed
- $e(t)$ is not observed $\Rightarrow H(z^{-1})$ is monic by appropriate scaling
- $H(z^{-1})$ and $H^{-1}(z^{-1})$ should be stable

271/316

A monic system is a system of which the impulse response starts with 1. Convince yourself that this is true for the systems $H(z^{-1})$ defined above, and also for $H^{-1}(z^{-1})$.

The fact that H^{-1} should be stable is because it will appear as a weight in the cost function, derived below.

7. Estimation parametric plant and/or noise models

7.a Classical time domain prediction error framework (cont'd)

• One-step-ahead prediction

$$v(t) = H(q)e(t) = e(t) + (H(q) - 1)e(t) = e(t) + (1 - H^{-1}(q))v(t)$$

One-step-ahead predictor process noise

$$\hat{v}(t|t-1) = (1 - H^{-1}(q))v(t)$$

= best (in least square sense) estimate of $v(t)$ based on the knowledge of all past samples $v(t-1), v(t-2), \dots, v(-\infty)$

One-step-ahead predictor output

$$\hat{y}(t|t-1) = G(q)u(t) + \hat{v}(t|t-1) = G(q)u(t) + (1 - H^{-1}(q))(y(t) - G(q)u(t))$$

Prediction error

$$\begin{aligned} \epsilon(t) &= y(t) - \hat{y}(t|t-1) \\ &= y(t) - G(q)u(t) - (1 - H^{-1}(q))(y(t) - G(q)u(t)) \\ &= H^{-1}(q)(y(t) - G(q)u(t)) \end{aligned}$$

272/316

Note that the one-step-ahead prediction $\hat{y}(t|t-1)$ uses the input u up to time t and the output y up to time $t-1$ (because the first element of the impulse response of $(1 - H^{-1}(q))$ is 0).

As such, the one-step-ahead prediction $\hat{y}(t|t-1)$ is the best prediction (in least squares sense) of $y(t)$ given the current input $u(t)$ and all the previous input and output values.

7. Estimation parametric plant and/or noise models

7.a Classical time domain prediction error framework (cont'd)

• Prediction error estimator

Prediction error cost function

$$V_{PE}(\theta, z) = \sum_{t=0}^{N-1} \varepsilon^2(t, \theta) = \sum_{t=0}^{N-1} (H^{-1}(q, \theta)(y(t) - G(q, \theta)u(t)))^2$$

Notes:

- the effect of the initial conditions is hidden in the filter operations
- the predictor should be stable, viz., H^{-1} and $H^{-1}G$
- according to the particular model structure one distinguishes

ARX: $C = 1$, and $D = A \Rightarrow$ linear least squares

ARMAX: $D = A$

ARMA: $G = 0 \Rightarrow$ time series/spectral analysis

OE: $H = 1$

BJ: G and H are independently parametrized

- frequency domain equivalent (Parseval's theorem)

$$\sum_{t=0}^{N-1} \varepsilon^2(t, \theta) = \sum_{k=0}^{N-1} |\varepsilon(z_k^{-1}, \theta)|^2$$

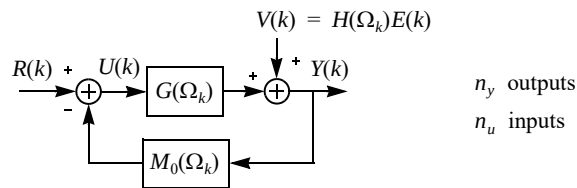
$$\varepsilon(z_k^{-1}, \theta) = H^{-1}(z_k^{-1}, \theta)(Y(k) - G(z_k^{-1}, \theta)U(k) - T_G(z_k, \theta) - T_H(z_k, \theta))$$

273/316

Recall the common notations of the system equation (ARX, ARMAX, ...) in Section 3.3.1.

7. Estimation parametric plant and/or noise models

7.g Multivariable systems in closed loop with known controller



• Gaussian ML estimator

$$-\sum_{k \in K} \log(f_{Y(k)}(Y(k)|R(k), \theta, \Lambda)) = C_1 + \sum_{k \in K} \log(|\det(S(\Omega_k, \theta))|^2) + F \log \det(\Lambda) + \dots$$

$$\sum_{k \in K} \varepsilon^H(\Omega_k, \theta) \Lambda^{-1} \varepsilon(\Omega_k, \theta)$$

with

$$\varepsilon(\Omega_k, \theta) = H^{-1}(\Omega_k, \theta)(Y(k) - G(\Omega_k, \theta)U(k))$$

$$S(\Omega_k, \theta) = (I_{n_y} + G(\Omega_k, \theta)M_0(\Omega_k))^{-1}H(\Omega_k, \theta) \quad \text{and} \quad \Lambda = E\{E(k)E^H(k)\} \in \mathbb{R}^{n_y \times n_y}$$

293/316

4.3 Taking into account the setpoint of the system

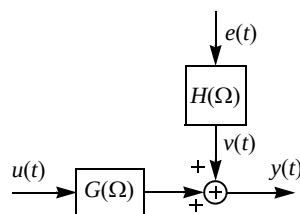
Linearisation around a setpoint, which needs to be determined.

Improved non-parametric methods

5.1 Local parametric methods

3. Improved nonparametric methods for LTI systems

3.a Arbitrary excitations — generalised output error



$$Y(k) = G(\Omega_k)U(k) + T(\Omega_k) + V(k)$$

$$V(k) = H(\Omega_k)E(k)$$

$$T(\Omega_k) = T_H(\Omega_k) + T_G(\Omega_k)$$

Goal: starting from $U(k)$ and $Y(k)$, obtain a nonparametric estimate of

- the frequency response function $G(\Omega)$
- the noise covariance matrix $C_V(k) = \text{Cov}(V(k))$

Difficulties: the presence of

- $T(\Omega_k)$ for estimating $G(\Omega_k)$
- $T(\Omega_k)$ and $G(\Omega_k)U(k)$ for estimating $C_V(k)$

Property:

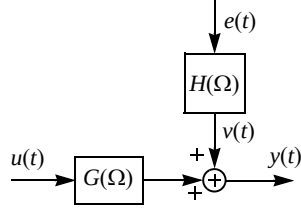
- $G(\Omega)$, $H(\Omega)$, and $T(\Omega)$ are smooth functions of the frequency

136/316

5.1.1 Local polynomial model

3. Improved nonparametric methods for LTI systems

3.a Arbitrary excitations — generalised output error (cont'd)



$$Y(k) = G(\Omega_k)U(k) + T(\Omega_k) + V(k)$$

$$V(k) = H(\Omega_k)E(k)$$

$$T(\Omega_k) = T_H(\Omega_k) + T_G(\Omega_k)$$

Basic idea solution: local polynomial approximation of $G(\Omega)$ and $T(\Omega)$

$$Y(k+r) = G(\Omega_{k+r})U(k+r) + T(\Omega_{k+r}) + V(k+r)$$

with Taylor series expansion

$$G(\Omega_{k+r}) = G(\Omega_k) + \sum_{s=1}^R g_s(k) r^s + O((r/N)^{(R+1)})$$

$$T(\Omega_{k+r}) = T(\Omega_k) + \sum_{s=1}^R t_s(k) r^s + N^{-1/2} O((r/N)^{(R+1)})$$

and

$$r = -n, -n+1, \dots, -1, 0, 1, \dots, n-1, n$$

$$\Theta = \begin{bmatrix} G(\Omega_k) & g_1(k) & g_2(k) & \dots & g_R(k) & T(\Omega_k) & t_1(k) & t_2(k) & \dots & t_R(k) \end{bmatrix}$$

137/316

5.1.2 Local rational model

Guidelines

6.1 Basic questions: Stochastic framework and Application

8. Guidelines

8.a Two basic questions

Stochastic framework?

- Output observations only
- Known input - output disturbed by unobserved input(s)

- Errors-in-variables
- Output error - indep. plant and noise dyn.

Noise model

- Parametric

- Nonparametric

Identification method

- Prediction error
- Frequency domain maximum likelihood

- Sample maximum likelihood

303/316

8. Guidelines

8.a Two basic questions (cont'd)

Application?	• Digital control	• Physical interpretation • Analog simulation	• Digital simulation
<i>Exp. setup</i>	• Zero-order-hold	• Band-limited	• Band-limited
<i>Domain</i>	• Discrete-time	• Continuous-time	• Continuous-time

304/316

6.2 Identification step by step

1. Check and selection of the experimental setup
2. Design of an experiment
3. Choice of the noise model
4. Preprocessing of the data
5. The identification step

8. Guidelines

8.b Identification step by step (cont'd)

Check and selection of the experimental setup

- Visit the site of the experimental setup and talk to the operators to learn from their experience
- Check the systematic errors of the complete data acquisition - calibrate the setup
- Check the validity of the signal assumptions (ZOH or BL, anti-alias filters)
- Pay attention to the synchronisation of the setup
- Always save the signal stored in the arbitrary waveform generator together with the noisy input-output data

306/316

8. Guidelines

8.b Identification step by step (cont'd)

Design of an experiment

- Choose the excitation (random or periodic) that best fits your specific needs (e.g. mimic the operational perturbations: same power spectrum and pdf)
- Select the amplitude range and frequency band of the excitation signal to cover the frequency band of interest
- Check for the presence of nonlinear distortions and the time-varying effects
- Check whether the device is captured in a feedback loop
- Keep your application in mind

307/316

8. Guidelines

8.b Identification step by step (cont'd)

Choice noise model

- Default choice: nonparametric models
- Use parametric noise models for the following problems:
 - output data only
 - known input / output disturbed by unobserved random input(s)

308/316

8. Guidelines

8.b Identification step by step (cont'd)

Preprocessing of the data

- Removal of trends, drifts, and offsets
 - Check for trends by calculating the mean value over the successive periods
 - Do not use the DC information during the identification
- Dealing with outliers and missing data
 - Perform new experiments
 - If this is not possible, estimate first nonparametrically the FRF and the missing data. Next, use this as starting value for the parametric estimation of the plant model in the presence of missing data
- Check whether the system is linear and time invariant
 - Check the presence of non-excited harmonics in the DFT spectra
 - Check the presence of skirts in the output DFT spectrum of all samples
- Estimate the nonparametric FRF
 - Calculate the FRF + noise level + level nonlinear distortions + level time-varying effects

309/316

... and select the frequency band of interest.

8. Guidelines

8.b Identification step by step (cont'd)

The identification step

- Choice of a model class
 - s -, z - or $\sqrt{s}$ - domain
- Model selection/ validation
 - Compare the transfer function model to the nonparametric FRF estimate
 - Analyse the value of the cost function
 - Perform a whiteness test of the residuals
 - Detect overmodeling via AIC (prediction) or MDL (physical interpretation)
- Impact of transient/leakage errors
 - Suppress nonparametrically the transient (leakage) errors in the data
 - If the transient time can be neglected, measure the steady state response
 - Always add the plant and noise transient terms when estimating parametric noise models
 - Suppress (non)parametrically the transient (leakage) errors in the validation data

311/316

8. Guidelines

8.b Identification step by step (cont'd)

The identification step

- Choice of a model class
 - s -, z - or $\sqrt{s}$ - domain
- Model selection/ validation
 - Compare the transfer function model to the nonparametric FRF estimate
 - Analyse the value of the cost function
 - Perform a whiteness test of the residuals
 - Detect overmodeling via AIC (prediction) or MDL (physical interpretation)
- Impact of transient/leakage errors
 - Suppress nonparametrically the transient (leakage) errors in the data
 - If the transient time can be neglected, measure the steady state response
 - Always add the plant and noise transient terms when estimating parametric noise models
 - Suppress (non)parametrically the transient (leakage) errors in the validation data

311/316

Appendix

Bibliography

- [1] R. Pintelon, P. Guillaume, and J. Schoukens. Uncertainty calculation in (operational) modal analysis. *Mechanical Systems and Signal Processing*, 21(6):2359 – 2373, 2007.
- [2] R. Pintelon and J. Schoukens. *System Identification: A Frequency Domain Approach*. John Wiley, 2nd edition, 2012.
- [3] Rik Pintelon and Johan Schoukens. FRF measurement of nonlinear systems operating in closed loop. *IEEE Transactions on Instrumentation and Measurement*, 62(5):1334–1345, 2013.
- [4] Johan Schoukens, Rik Pintelon, and Yves Rolain. *Mastering System Identification in 100 Exercises*. IEEE Press, John Wiley & Sons, 2012.